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MORGAN STANLEY RESEARCH
ASIA/PACIFIC

Qing Wang

Qing.Wang@morganstanley.com
+852 2848-5220
Morgan Stanley Asia Limited+

Steven Zhang

Steven.Zhang@morganstanley.com
+852 2326-0029
Morgan Stanley Asia Limited+

Ernest Ho

Ernest.Ho@morganstanley.com
+852 2239-7818
Morgan Stanley Asia Limited+

The China Files Chinese Economy through 2020

As part of our *China Files* series, this report lays out the macroeconomic context.

We examine the likely deceleration in growth over the next decade, what demographic shifts suggest for China's labor supply, and three factors that herald good news for Chinese consumers. Companion pieces in the ongoing *China Files* series identify global corporations best positioned for the opportunities in China over the next five years.

Not whether but how growth will decelerate. China's growth will likely decelerate, but through 2020 we can still expect to see average gains of 8% per year as overall consumption, service sector spending, and income rise relative to GDP.

Labor supply to remain abundant. China will continue to benefit from a low demographic dependency ratio and abundant labor supply through 2020. The deceleration in the growth of the working-age population is unlikely to stall China's overall economic expansion.

A golden age for consumption. Consumption should become a key driver of economic growth. By 2020, we expect Chinese consumption will be at two-thirds of the US level and account for about 12% of the world total. China's private consumption appears to be substantially underestimated.

Also in the *China Files* series:

The China Files: US Corporates and China's Megatransition, September 20, 2010

The China Files: European Corporates & China's Megatransition, October 29, 2010

The China Files: Asian Corporates & China's Megatransition, November 8, 2010

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Introduction

By now, the story of China's rapid economic acceleration is the stuff of everyday news. A low-cost labor base. The world's largest exporter. Favorable exchange rates. Double-digit GDP growth (despite the global recession). And the most rapidly expanding economy in the world.

What's less often discussed, however, is how China came to occupy this place of increasing importance in the global marketplace—and what China's rapid economic acceleration might portend for new and existing businesses around the world.

In this report, authors Qing Wang, Steven Zhang, and Ernest Ho look forward to 2020 and address three topics of increasing interest to observers of China's economic evolution: the likely deceleration in China's growth over the next decade, what certain demographic shifts suggest for China's labor supply, and the three factors that herald good news for Chinese consumers.

The ongoing *China Files* series also includes reports with more detailed reviews of global corporations that are positioning themselves for opportunities in China over the next five years. Teams of analysts covering more than 20 separate industries bring together our experts from China with their colleagues covering the same industries in other regions. Together they look at the determining factors for success in the region and identify which companies are likely to benefit from the secular drivers—or megatrends—that are pushing China toward one of the biggest economic transformations in its history—the megatransition.

Defining and Assessing the Megatrends

In the first section of this report, *Not Whether but How Growth Will Decelerate*, our team of China economists shares its view on China's growth outlook and the structural evolution of the economy over the next decade. They contend that China's growth will likely decelerate, but that through 2020 we can still expect to see average gains of 8% per year as overall consumption, service sector spending, and income rise relative to GDP.

In the second section, *Labor Supply to Remain Abundant*, the authors assess how China's labor force will evolve and the implications for economic growth over the next decade. They expect China will continue to benefit from a low demographic dependency ratio and abundant labor supply through 2020. The deceleration in the growth of the working-age population is unlikely to stall China's overall economic expansion.

In the third and final section, *A Golden Age For Consumption*, they predict that consumption in China will take off over the next decade as a driver of growth.

- The Chinese economy is at an inflection point. By 2020 Chinese consumption will be at two-thirds of the US level and account for about 12% of the world total.
- Three sources of consumption growth—rising income, lower savings ratio, and consumption upgrades—suggest a “golden age of consumption” is about to occur in China.
- Large regional disparities make China special. A framework for describing regional dynamics helps us understand the evolution of China's growing rate of consumption.
- China's private consumption is substantially underestimated.

The China Files: An Ongoing Series...

As the China story unfolds, we will continue to address the investor debates surrounding one of the most intriguing economic evolutions of the decade.

Not Whether but How Growth Will Decelerate

A consensus is emerging that China's growth over the next decade could be lower than in the previous. We share the consensus view.

Growth is a function of labor, capital, and total factor productivity. China managed to maintain an average annual growth rate of 9.8% between 1978 and 2009. There are a number of studies in academic literature that attempt to estimate the contribution of capital, labor, and TFP to China's GDP growth. According to these studies, capital accumulation has contributed the most to growth (3.2 to 5.1 percentage points), and labor the least (0.7 to 2.8 percentage points). There is broad consensus in the literature that China's TFP growth has been quite strong, with the magnitude of its contribution to growth between 3.5 to 3.8 percentage points.

Looking ahead, these three factors are likely to decelerate over the coming decade.

Demographics imply slowing labor supply growth.

According to forecasts made by the United Nations, the average growth rate of the working-age population over the next decade will be about 0.23%, which is a rather sharp slowdown from the average growth rate of 1.28% over the past two decades. To the extent that there is a relatively stable relationship between the working age population and employment, this suggests that labor supply growth will slow accordingly.

Lower savings ratio, less capital formation. As China's population ages, the share of savers (aged 35-59) in the population should decline and that of dis-savers (aged 65 and above) rise; these divergent trends would lower the average household savings ratio. We also expect the government to step up its efforts to ask large, profitable state-owned enterprises to pay dividends to the state, which would reduce corporate savings. To the extent that government savings will not increase as a result of collecting more dividends from SOEs, domestic saving will decline. A lower savings ratio would contribute to a lower investment ratio and thus less capital formation.

Improvement in TFP comes from two main sources:

- *The reallocation of labor from low-productivity activity (such as farming in rural areas) to high-productivity activity (like manufacturing in urban areas).* In view of the substantial progress already made by China in moving surplus labor from rural to urban areas over the past three decades,

potential marginal TFP gains stemming from the latter will diminish, in our view.

- *Technological innovation and structural adjustment.* While the Chinese economy may continue to benefit from technological innovations over the next decade, potential TFP gains from structural reform are unlikely to be as strong as over the previous three decades. This is because the most obvious distortions—the low-hanging fruit—in the economy have already been largely removed during those three decades of structural reform. As economic development reaches a higher level, it becomes increasingly difficult to push through more profound reforms to address deep-seated structural rigidities, especially as private interests become more entrenched.

There is, however, much uncertainty over how fast the deceleration could turn out to be. There are several mitigating factors at work.

- *Potential agricultural land reform and deregulation of the household registration system.* These changes could free more labor from agricultural production and accelerate the rural-urban labor migration, which would effectively increase labor supply. This of course hinges on potential policy changes.
- *The investment to GDP ratio is likely to edge down only gradually.* This is because the existing pattern of growth and resource allocation has a strong momentum and rebalancing measures have so far been modest. China's national savings still exceed national investment, as is reflected in persistent and sizable current account surpluses. In this context, savings or availability of low-cost capital are unlikely to become an effective binding constraint for capital formation at least for another five years, in our view.
- *More difficult reforms could be carried out.* While the “low-hanging-fruit” reforms have now been largely completed, difficult reforms, if carried out, could potentially generate substantial efficiency and productivity gains. These reforms include improved financial intermediation by developing the bond market and/or deregulating interest rates, improving market access for private investors to sectors that are still subject to state monopoly (e.g., health care, education, telecommunication, media, and transportation).

A recent study by the World Bank has made a useful attempt at forecasting China's potential growth over the coming decade. According to WB estimate, the potential GDP growth would be 8.4% over 2011-15 and 7.0% over 2016-20. This

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estimate of potential GDP growth is broadly in line with or lower than the projections made in some earlier studies, which put the potential GDP growth in the range of 7.5%-9.6% over the next decade.¹

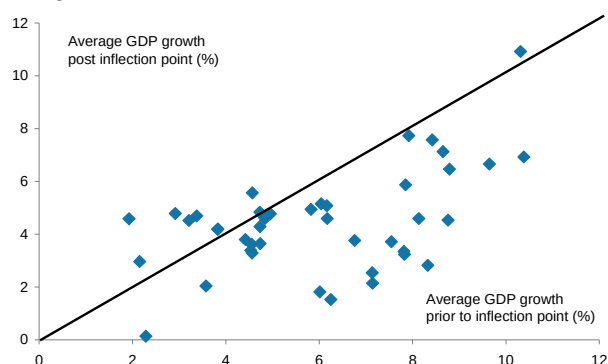
China's Potential Growth to Slow: Historical Perspective

World economic history also suggests that an economy tends to decelerate after reaching a certain level of development. This pattern can be established based on the comprehensive data set compiled by Angus Maddison, the author of the book *The World Economy: A Millennial Perspective*².

Specifically, we find that when GDP per capita for an economy reaches \$7,000, average GDP growth rates over the following decade tend to decelerate rather significantly³. According to Madison's data, about 40 economies have managed to reach GDP per capita of \$7,000 over the past 100 years or so. The growth rates of 31 out of these 40 economies slowed in the years after this particular income level was reached (exhibit 1). The average growth rate over the following 10 years after the inflection point was achieved was 2.8 percentage points less than the average growth rate over the 10 years in the runup to the inflection point.

Exhibit 1

History: Growth Deceleration Post Inflection Point



Sources: Company data, Morgan Stanley Research.

Based on Madison's data, we estimate that China's GDP per capita reached \$7,000 around 2008. If history is a guide and

¹ Xiaolu Wang, Gang Fan, and Peng Liu, 2007, "Pattern and Sustainability of China's Economic Growth towards 2020"; He Jianwu, Li Shantong, and Sandra Polaski, April 2007, "China's Economic Prospects 2006-2020"; OECD, 2005, "OECD Economic Review - China"; World Bank, June 2010, "China Through 2020 - A Macroeconomic Scenario."

² Angus Maddison, 2006, *The World Economy: A Millennial Perspective*, OECD.

³ The US dollar here is defined in Maddison's study as "Geary-Khamis dollar" (GK\$). GK\$, a hypothetical unit of currency, has the same purchasing power as US dollar had in the US at a given time. GK\$ denomination allows consistent international comparison across countries.

the law of gravity applies to China, China's economic growth is set to slow.

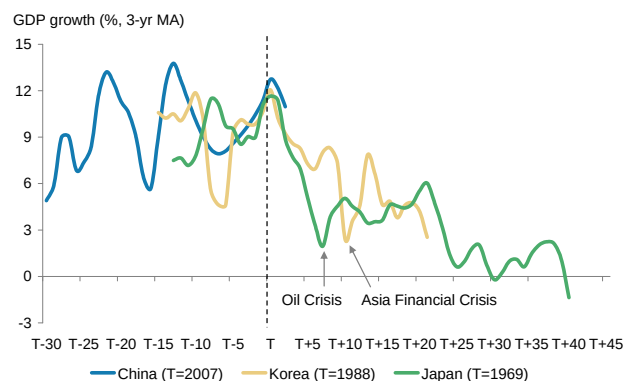
How Growth Deceleration Could Pan Out: The Experiences of Japan and Korea

If deceleration in growth is not subject to debate, the next key issue is how the deceleration is likely to pan out. In this regard, the experiences of Japan and Korea provide a particularly useful reference, in our view. China's economic success is not unique and is part of the "Asian economic miracle." China's economic development model is still largely a replication of that adopted by other more advanced East Asian economies, in our view.

China is at a similar inflection point. According to the data compiled by Maddison, Japan and Korea reached the \$7,000 level in the late 1960s and 1980s, respectively. Thus the Chinese economy is now at an inflection point similar to that in the Japanese economy 40 years ago (i.e., around 1969) and the Korean economy 20 years ago (i.e., around 1988). We find that beyond this inflection point, overall GDP growth tended to decelerate and inflation to accelerate in both Japan and Korea (exhibits 2 and 3).

Exhibit 2

Growth Deceleration Post Inflection Point

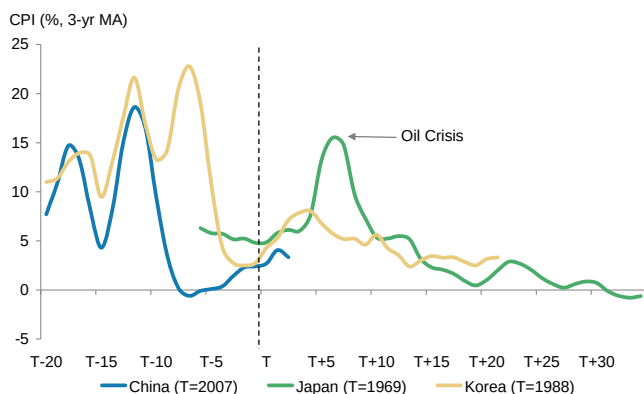


Sources: CEIC, Morgan Stanley Research.

Specifically, with regard to economic growth, while the 10-year average GDP growth rate for Japan during 1960-69 and Korea during 1979-88 was 10.4% and 10.0%, respectively, the rate in 1970-79 and 1989-1998 decelerated to 5.2% and 6.3%, respectively (exhibit 2). With regard to inflation, while the 10-year average CPI inflation rates for Japan during 1960-69 and eight-year average inflation for Korea during 1982-88 were 5.4% and 4.0%, respectively, their 10-year average CPI inflation rates during 1969-74 and 1989-93 accelerated to 12.4% and 6.2%, respectively (exhibit 3).

Exhibit 3

Inflation Acceleration Post Inflection Point



Sources: CEIC, Morgan Stanley Research.

It should be pointed out that the comparison is distorted by two “accidents”:

- The oil crisis in the mid 1970s. This caused Japan's GDP growth to slump and inflation to surge.
- The Asian financial crisis in late 1990s. This explained the sharp slowdown in Korea's growth.

Excluding the impact of these shocks, the economic deceleration and inflation acceleration would have been less dramatic: The average growth rates post the inflection point would have declined by 4.1 (instead of 5.2) percentage points for Japan and 2.4 (instead of 3.7) percentage points for Korea, and Japan's average CPI inflation after the inflection point would have been 6.3% (instead of 12.6%).

Profound Structural Transformation: The Experiences of Japan and Korea

Underlying the important developments of GDP growth and CPI inflation at the aggregate level, the economic structure of both Japan and Korea has also undergone profound transformation before and after the inflection point was crossed (exhibits 4 through 8).

There are three aspects of the economic structure:

- Production (i.e., agriculture, industry, and service);
- Expenditure (i.e., consumption, investment, and net exports),
- Income (i.e., labor compensation, profits, and government net revenue).

The structure of both the Japanese and Korean economies began to go through quite meaningful changes after the inflection point was crossed. In particular, the three key ratios

of the economy—consumption-GDP, service sector-GDP, and labor income-GDP—tended to rise rather rapidly.

With regard to production structure, the ratio of the service sector to GDP increased from 50% in 1969 to 58% in 1979 in Japan and from 44% in 1988 to 52% in 1998 in Korea.

With regard to expenditure structure, the ratio of consumption to GDP increased from 60% to 69% in Japan and from 60% to 65% in Korea.

With regard to income structure, the ratio of labor compensation to GDP ratio increased from 43% to 54% in Japan and from 41% to 44% in Korea (exhibit 4).

Exhibit 4

Japan and Korea: Profound Structural Transformation amid Growth Deceleration

	Japan (1969)		Korea (1988)	
	Before	After	Before	After
Macro				
GDP growth (%)	10.4	6.3	10.0	7.6
CPI Inflation (%)	5.4	6.9	4.0	6.2
Structure (in percent of GDP)				
Production				
Agriculture	11	10	25	22
Manufacturing	39	32	30	26
Service	50	58	44	52
Expenditure				
Consumption	60	69	60	65
Investment	39	32	38	35
Income				
Labor	43	54	41	44
Profits	38	27	37	31

Note: For GDP and CPI, 10-year average before and after the inflection points were calculated, respectively, with the outlier data points—1969 for Japan and 1988 for Korea—during crisis excluded. For indicators of economic structure, the ratio under “Before” refers to the data point at inflection point and that under “After” refers to the data point at end of the decade after inflection point is reached.

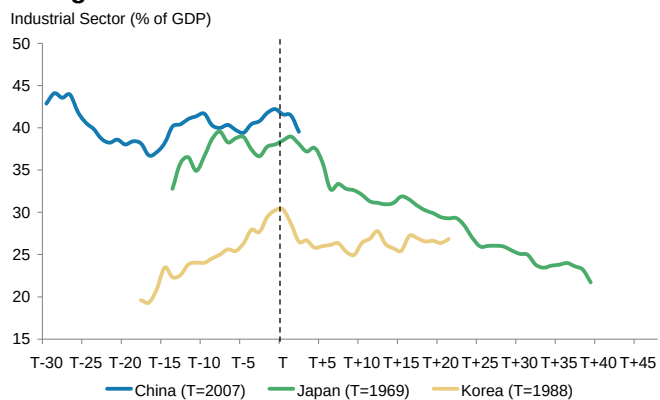
Sources: CEIC, Morgan Stanley Research.

While these structural transformations did not evolve in a straight-line fashion, the trends after the inflection point was crossed were clear and pronounced (exhibits 5 through 8). And the structural change in Japan has been more rapid than that in Korea.

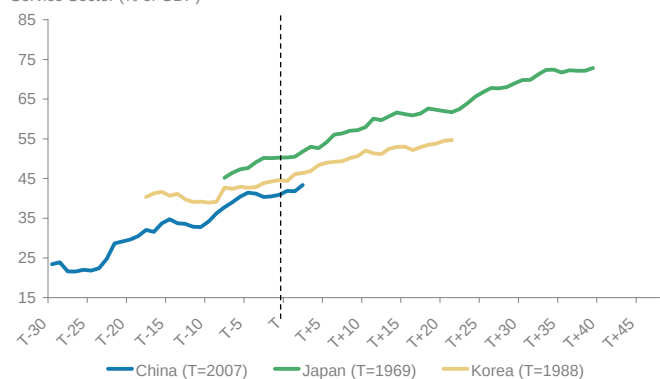
However, of particular note, the trends for net exports are less clear cut: While Korea experienced a sharp reduction of trade surplus post inflection point, Japan's trade surpluses remained large and even widened and were only temporarily reduced by the oil crisis in mid-1970s (exhibit 7).

Exhibit 5

Rising Share of Service Sector Post Inflection Point



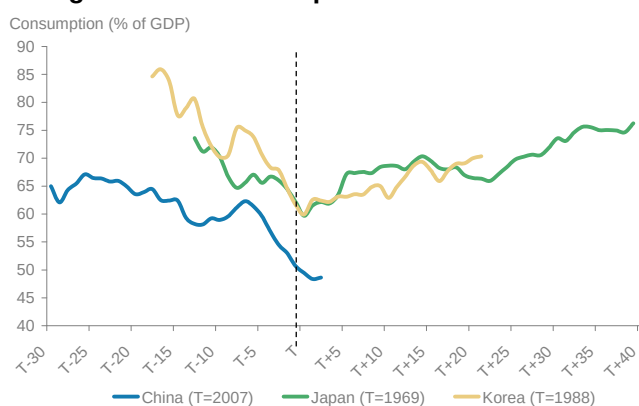
Service Sector (% of GDP)



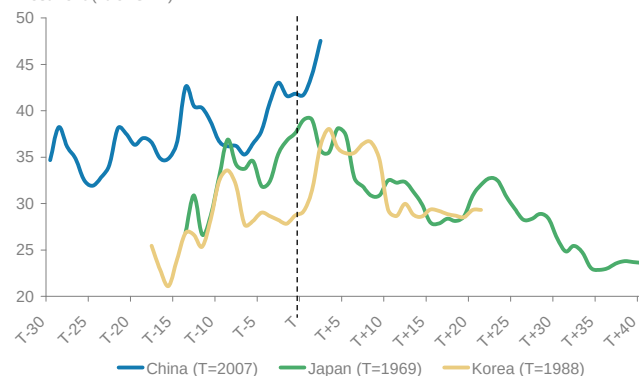
Sources: CEIC, Morgan Stanley Research.

Exhibit 6

Rising Share of Consumption Post Inflection Point



Investment (% of GDP)



Sources: CEIC, Morgan Stanley Research.

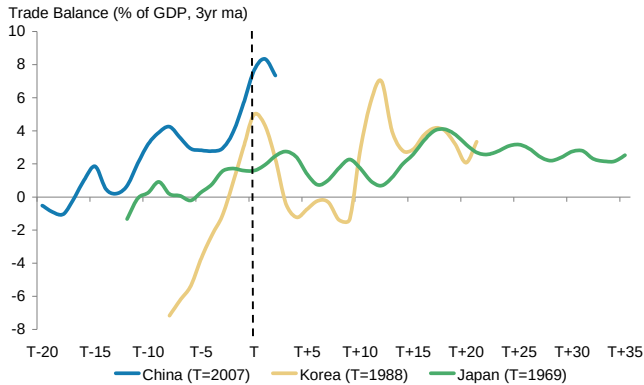
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Exhibit 7

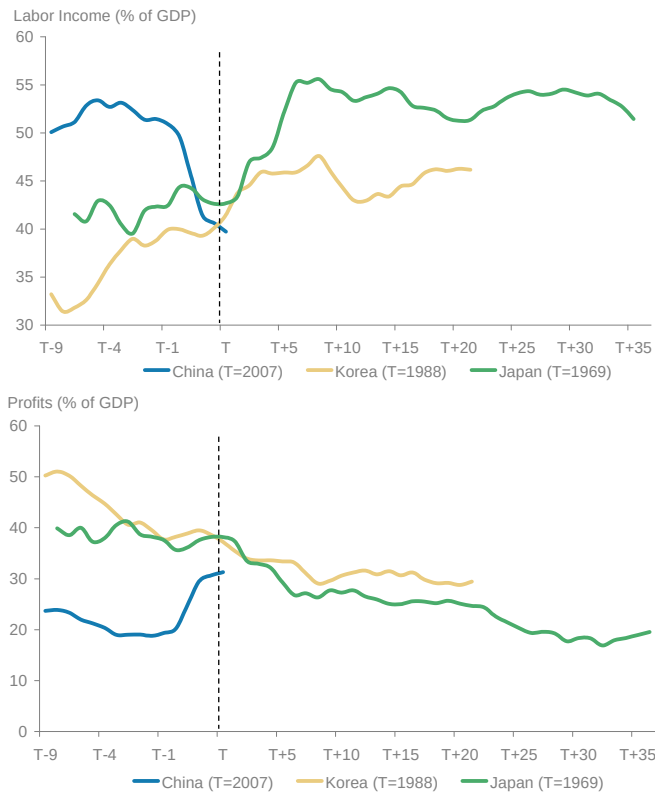
No Clear Trend for Net Exports



Sources: CEIC, Morgan Stanley Research.

Exhibit 8

Rising Share of Labor Income Post Inflation Point



Sources: CEIC, Morgan Stanley Research.

Growth Deceleration from High Levels: Sectoral Evidence

The experiences of both Japan and Korea suggest that economic growth rates will decelerate after an economy reaches \$7,000 GDP per capita (based on Maddison's data). While the decline in the average growth rate is significant, it

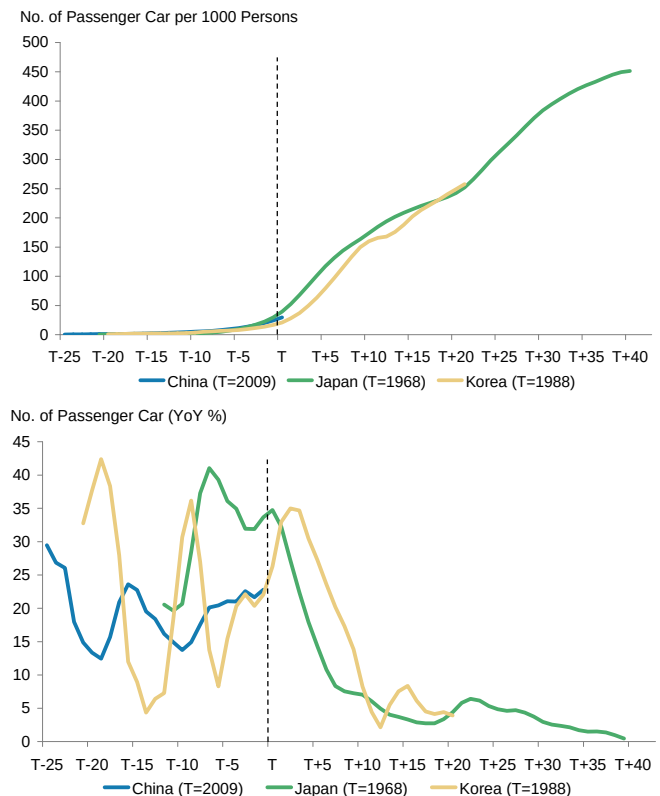
should be stressed that growth decelerated from a high level. Average growth rates after the inflection point were still quite robust, allowing sustained, decent improvement in living standards.

This point can be further highlighted by such sectoral data as power usage, steel consumption, and auto ownership. As shown in exhibits 9 through 11, even after the inflection point was crossed, both Japan and Korea continued to improve their level of development quite decently, as reflected in rising consumption of electricity and steel, as well as auto ownership. However, these upward trends should not be confused with the trends of growth rates, which decelerated quite significantly during the same period of time.

Auto ownership is a case in point (exhibit 9). During the 10 years after the inflection point, the number of passenger cars per 1,000 persons increased by 3.5 times in Japan and 6.5 times in Korea! Nevertheless, their respective growth rates both registered sharp deceleration during the same period of time. This in part reflected the fact that the sharp acceleration in growth in the run-up to the inflection point was from very low levels.

Exhibit 9

Number of Passenger Cars: Level vs. Growth



Sources: CEIC, Morgan Stanley Research.

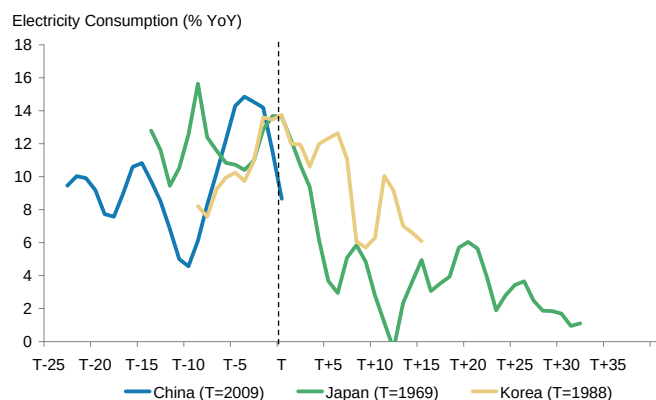
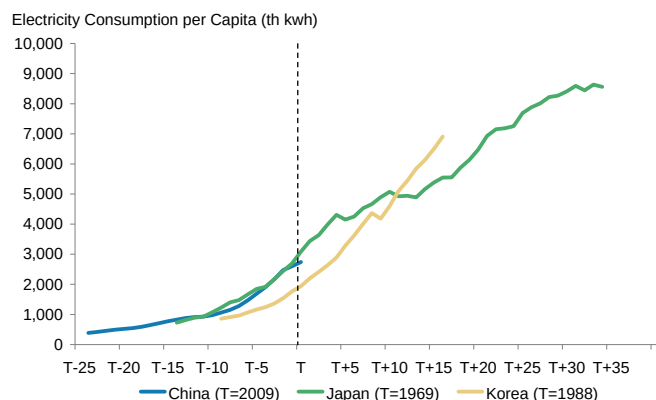
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Exhibit 10

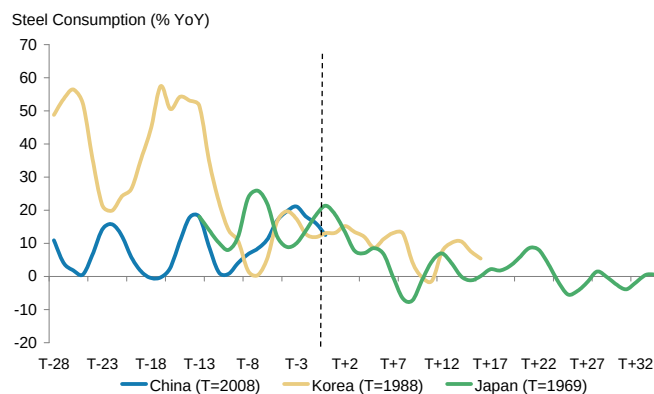
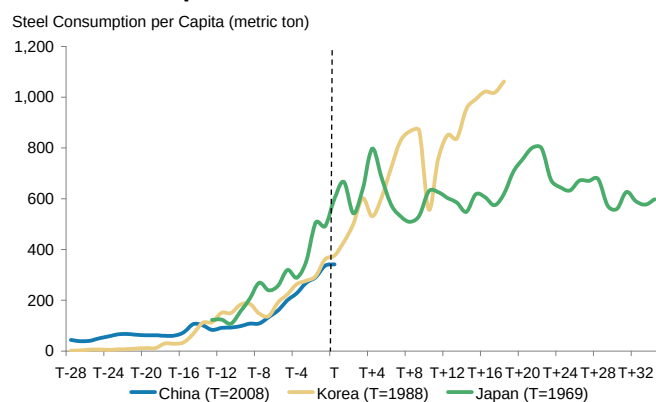
Electricity Power Consumption: Level vs. Growth



Sources: CEIC, Morgan Stanley Research.

Exhibit 11

Steel Consumption: Level vs. Growth



Sources: CEIC, Morgan Stanley Research..

China Is Special After All: The Regional Dimension

China's successful economic development is not unique and is actually part of the "Asian economic miracle" pioneered by Japan and followed by other East Asian countries. Since, in terms of economic development level, China is at a stage similar to that of Japan 40 years ago and Korea 20 years ago, the developmental experiences of Japan since the late 1960s and Korea since the late 1980s are particularly relevant in helping to assess China's potential growth trajectory over the next decade, in our view. They set a powerful benchmark for gauging the potential trends for the Chinese economy over the next decade.

Several key takeaways:

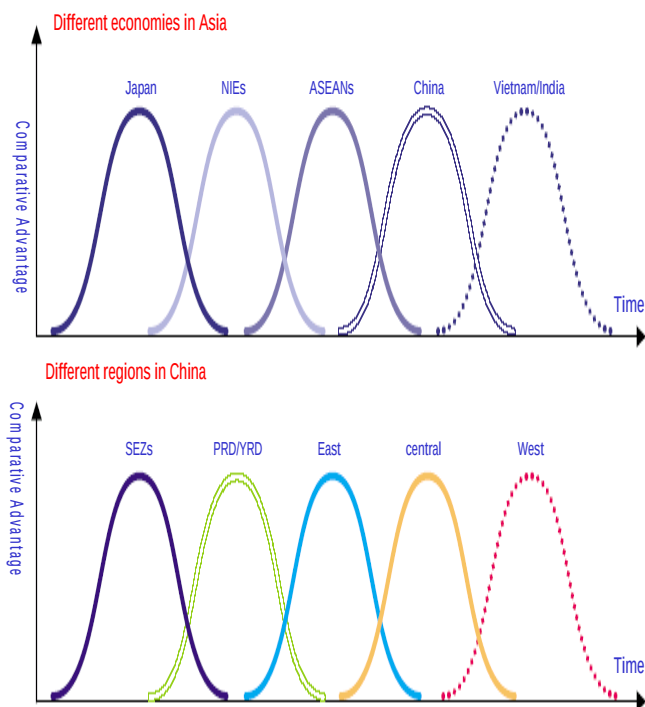
- Meaningful growth deceleration will be inevitable and the level of sustained inflation will be higher.
- Consumption and the service sector will outpace overall economic growth.
- Labor will gain bargaining power over time and take a bigger share of the national income.
- There is no clear trend for trade balance.

Moreover, if China were to follow the same growth trajectory as Japan and Korea after their economies crossed the inflection point, China's average economic growth over the next decade would be in the range of 6.0%-8.0%, or a decline of 2 (i.e., Korea's case) to 4 (i.e., Japan's case) percentage points from the average 10% GDP growth over the past 10 years.

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Exhibit 12

Flying Geese Paradigm: China Emulates East Asia

Source: Morgan Stanley Research.

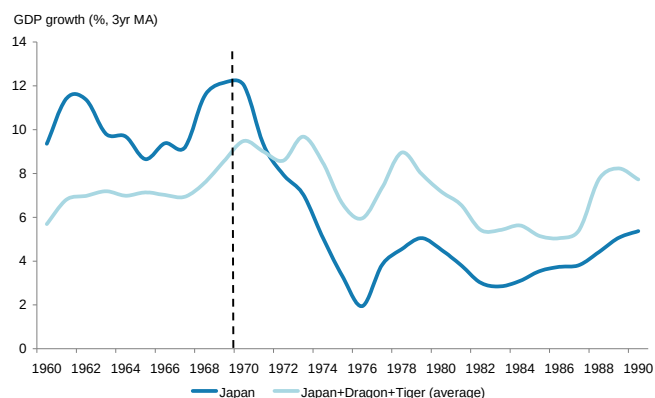
Indeed, we cannot rule out the possibility that the Chinese economy may follow the exact same growth path as those of Japan and Korea. However, as a continental economy and given its vast regional disparity, we believe China's economic evolution is likely to be more gradual than was that of its East Asian peers. Here is why:

"Asia's economic miracle" is best illustrated by the "Flying Geese Paradigm" (exhibit 12). This theory postulates that Asian nations will catch up with more advanced economies through a regional hierarchy where the production of commoditized goods moves continuously from the more- to the less-advanced countries. The lead goose in this pattern is Japan, followed by the new industrializing economies (South Korea, Taiwan, Singapore, and Hong Kong). After these two "geese" come the main ASEAN countries: Philippines, Indonesia, Thailand, and Malaysia. Finally, the least-developed major nations in the region—China, Vietnam, and India—would make up the rear guard in the formation.

We argue that, given the sheer size of the Chinese economy and population, and the country's vast regional disparity, China's internal development would emulate the "Flying

Geese Paradigm," where the lead goose is the special economic zones, followed by the second-tier regions (the Pearl River and Yangtze River Deltas). After the second-tier regions would come the other eastern provinces, with the central and western provinces bringing up the rear (exhibit 12). (See *China Economics: One Country, Three Economies: Play the Regional Disparity in China*, May 26, 2009.)

Exhibit 13

The Flying Geese As a Whole Decelerated Less

Sources: Madison (2006), Morgan Stanley Research.

Under this framework, we believe the economic transformation experiences of individual Eastern economies like Japan and Korea are likely to serve as a useful reference for a potential adverse-case scenario for Chinese economy. Put differently, while the Chinese economy is likely to decelerate as the Japanese and Korean economies did, the pace of deceleration in China may not be as fast, and the complete transition from high-growth to medium-growth will take longer than it did in Japan and Korea.

This argument is consistent with the growth experiences in East Asia. The average growth rate of Japan, the four Asian dragons (Korea, Taiwan, Singapore, and Hong Kong), and the four Asian tigers (Malaysia, Thailand, Indonesia, and Philippines) is faster than the growth rate of Japan alone, after the inflection point was crossed (exhibit 13). In another words, the growth deceleration for the "flying geese" as a whole is much milder than that of the leading goose (i.e., Japan).

Chinese Economy through 2020

Based on the analysis discussed above, we have come up with the broad parameters for the economy through 2020 under a base case and two alternative scenarios (exhibit 14).

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Exhibit 14

Chinese Economy through 2020: Three Scenarios

	2009-09	2010-20		
		Baseline	Alt I	Alt II
Probability (%)		70	20	10
Macro				
GDP growth (%)	10.3	8.0	9.5	6.5
CPI Inflation (%)	1.9	3.5	2.5	4.0
Structure				
Production				
Agriculture	17	15	15	15
Manufacturing	40	35	40	33
Service	43	50	45	52
Expenditure				
Consumption	49	54	47	58
Investment	48	43	50	41
Income				
Labor	40	45	40	52
Profits	31	30	35	20

Note: For GDP and CPI, 10-year average before and after the inflection points were calculated, respectively.

Source: Morgan Stanley Research.

Under the base case scenario, we believe the Chinese economy will still be able to maintain an average of 8% growth per annum over the next decade. This scenario implies a slowdown of slightly more than 2 percentage points from the average growth achieved in the previous decade. We expect the deceleration in China's growth to be slower than that displayed by either Japan or Korea. Average CPI inflation over the next decade would be 3.5%, a significant acceleration from the average inflation rate of 1.9% over the previous decade. The higher inflation rate mainly reflects a labor market that is increasingly normalized.

At this pace of development, China's nominal GDP would triple from its current size, reaching Rmb103 trillion. Assuming an average 3% appreciation of the renminbi against the US dollar per annum, Chinese nominal GDP in US dollar terms would quadruple its current size, reaching US\$20 trillion by 2020.

We envisage the economic structure undergoing meaningful changes. These, however, are less drastic than those experienced by Japan and Korea.

More consumption, less investment relative to GDP.

Specifically, we expect the consumption-GDP ratio to rise by 5 percentage points by the end of the next decade, while the investment-GDP ratio will decline by 5 percentage points by end of the next decade. This would entail average consumption growth to outpace and investment growth to underperform the overall GDP growth by roughly 0.5

percentage points per annum, respectively. We expect the trade surplus to be broadly stable at about 3% of GDP.

Labor's share of GDP to rise, government's to fall. Another notable change that we foresee over the next decade is that labor compensation will account for a large share of national income. This rising share of labor will be largely offset by a corresponding decline of the share of government, with a modest decline in the share of profits. This is because the current share of government is already quite high, especially compared to that of Japan and Korea, and there is room for tax cuts.

Industrial sector's share declines, service's share rises. We expect the industrial sector-to-GDP ratio to decline by 5 percentage points, offset by a rising share of services. Nevertheless, the industrial share is still relatively high, at 35% by 2020.

We construct two alternative scenarios

Alternative I. Current trends continue (20% probability). In this first scenario, the Chinese economy would basically continue business as usual, with no material change compared to the previous decade, featuring strong growth, modest inflation, and no meaningful transformation of the economic structure. If this scenario were to materialize, the Chinese economy would be proven to be unique, defying the development patterns of Japan and Korea. We assign 20% subjective probability to this scenario.

This scenario would be possible if:

- The global economy were to stage a strong recovery back to pre-crisis levels and sustain its pace of expansion thereafter; or
- China's economy were to prove to be so competitive as to successfully penetrate other emerging markets such that China's export growth would be as strong as pre-crisis, without causing backlash of trade protectionism measures, despite further expansion of market share.

Alternative II. A Japanese-style adjustment (10% probability). In this second scenario, the Chinese economy is envisaged to go through as drastic growth deceleration and structural adjustment as Japan did. Such a scenario could be catalyzed by either very proactive (or even draconian) policy intervention to correct the structure of the economy artificially or by external shocks such as a complete meltdown in external demand and sustained surge in international commodities prices due to supply shocks. Given Japan's experience, we are not able to rule out the possibility of this

scenario outright. However, we only assign a 10% subjective probability to this scenario.

Risks

Both Japanese and Korean economies were hit by serious crises in the decade after their economies crossed the inflection point. These experiences suggest that the future development for the Chinese economy over the coming decade will probably not be entirely “accident-free” (see *China Economics: China Macro Risk Radar*, September 3, 2010).

The particular crisis experiences of Japan and Korea point to the vulnerability of the Chinese economy and the types of possible risks that could seriously destabilize the economy. There are two—an energy crisis or a financial crisis.

In view of China's rising dependency on crude oil over the coming decade, a sharp increase in crude oil prices for a prolonged period of time could deal as large a negative impact on the Chinese economy as was the case for Japan during the oil crisis in the mid-1970s. In fact, the oil crisis in the mid-1970s caused the Japanese economy to gear down on a permanent basis.

Korea was hit hard by the Asian financial crisis in 1997-98. Current account deficits, coupled with an inflexible exchange rate arrangement, made the Korean economy vulnerable to a

sudden potential reversal of capital inflows. Although we expect that China will continue to run a significant trade surplus over the next decade, the situation could deteriorate sharply if there were a major investment boom funded by rapid credit expansion, which could cause a sharp increase in imports and turn China's trade surplus into a trade deficit. If, at the same time, the renminbi exchange rate were to remain inflexible, China could be exposed to the same types of shocks that caused the Asian financial crisis.

Investment Implications

We identify these megatrends over the next decade. For capital markets investors, riding these megatrends would mean positioning for:

- A secular boom in consumption underpinned by the rising power of labor;
- Rapid expansion of the service sector; and
- A deceleration in the traditional manufacturing sectors, despite relatively high growth and a shrinking share of profits.

Moreover, the central and western regions in China are expected to grow much faster than eastern regions.

Labor Supply to Remain Abundant

The debate over China's demographic transition and the attendant implications for the labor supply has been in the spotlight since 2003-04, when the shortage of rural migrant workers initially emerged. What were at first just concerns are now becoming real threats as the demographic total dependency ratio is poised to bottom and working-age population growth to decelerate over the coming years. Indeed, the expectation of a precipitate decline in the growth rate of the working-age population appears to have become one of the most powerful arguments for a sharp slowdown in China's economic growth over the next decade.

A deceleration in the growth of the working-age population is unlikely to be a headwind to overall economic expansion in China, making an average GDP growth of 8% per annum through 2020 quite achievable, in our view.

- While China's working-age population is set to decelerate significantly over the next decade, the effective labor supply will remain abundant. Several long-standing distortions in the labor market have driven a wedge between the working-age population and the effective labor supply, as shown both by the distinct disconnect between GDP and employment/working-age population growth over the past two decades and the current disproportionately high share of the rural population and employment in the primary sector.
- The removal of these distortions—which is taking place in a profound fashion—should ensure an abundant labor supply through to 2020. We forecast that while the working-age population is expected to increase only by 20 million, urban and rural non-farm employment will increase by 80 million from 2010-20.
- In the meantime, strengthened human capital investment and continued capital deepening—the other two key sources of labor productivity gains—are likely to persist in China.
- More specifically, given the high percentage of the working-age population living in rural areas and the disproportionately high share of primary sector employment, a significant boost in labor supply and labor productivity is, in our view, eminently achievable, especially with proper policies in place. Some such policies have already been implemented in recent years, including the gradual deregulation of controls over rural-urban labor mobility that have been impeding such mobility for over half a century, the acceleration of urbanization, the “Go West” inland investment program, an aggressive social housing program,

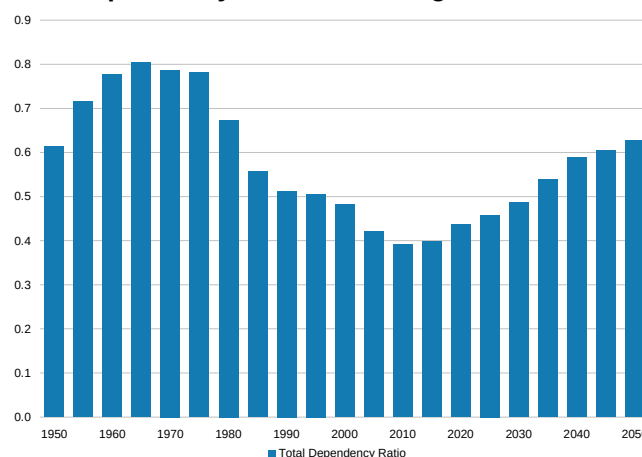
and the improved provision of education. Moreover, China should continue to be one of the most important manufacturing bases in the world, as it remains internationally competitive given its relatively high labor productivity, strong logistic infrastructure, and economies of scale.

What Is the Recent Debate?

China's demographic transition has been at the center of debate since 2003-04, when a shortage of rural migrant workers became evident in the Pearl River Delta. This was later reported to have spread to the whole country. It called into question the sustainability of the business model adopted by the “world factory,” which relies on a seemingly unlimited supply of migrant workers who, at very low cost, churn out more and more goods for export. The subsequent increase in wages was considered strong evidence of labor shortages, arousing concerns that China has already reached or crossed the “Lewis turning point” and, as a result, the era of unlimited labor supply would come to an abrupt end.

Exhibit 15

Total Dependency Ratio Bottoming Out



Sources: UN World Population Prospects (2008 Revision), Morgan Stanley Research.

These concerns appear to be becoming a real threat, as the total dependency ratio (the number of dependents, such as children and the elderly, compared to the working-age population) is poised to bottom out in the next few years, according to United Nation forecasts. This drop in the dependency ratio casts serious doubts on China's ability to remain the dominant manufacturing base (exhibit 15). Moreover, the continual increase in wages, coupled with the much-anticipated appreciation of the renminbi, are additional

hurdles, squeezing an already thin margin and hurting Chinese exporters' international competitiveness.

The pessimism is so widespread and deep-rooted in the minds of some China observers that an upbeat comment from the Minister of Commerce Chen Deming in March 2010 was met with considerable skepticism. Chen was quoted as saying that China could still enjoy another 10 years of "demographic dividends," based on the analysis done by the Ministry of Commerce, specifically on the labor shortage issue. It is our understanding that a key working assumption made by Chinese authorities in the preparation of their twelfth five-year plan (2011-15) is that demographics will continue to work in China's favor over the next decade.

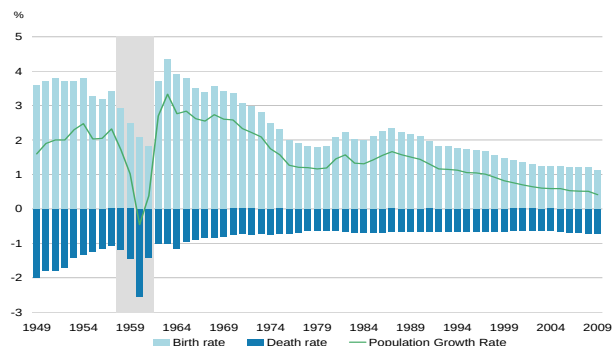
What causes the disconnect? Will demographics in general and labor supply in particular be a headwind or a tailwind to economic growth? This is a key issue that informs China's economic outlook through 2020.

How Did We Get to Where We Are Today?

China's demographic profile has changed a great deal since the establishment of the People's Republic of China in 1949. During the first two decades, the return of peace brought a period of high birth and declining mortality rates, laying the foundation for rapid population growth (exhibit 16). According to the National Bureau of Statistics of China, except for 1958-61, when the Great Leap Forward led to a temporary disruption in the trend (shaded area in exhibit 2), the birth rate remained elevated, at about 30 births per 1,000 people, while the death rate declined steadily, from 20 deaths per 1,000 people in 1949 to just eight in 1970. By 1970, the population consisted predominately of people of working-age⁴ and children under 14, while the elderly accounted for only 4% of the total population (exhibit 17). Growing by over 2% annually, the population reached 830 million in 1970, more than 50% above 1949's figure of 540 million.

Exhibit 16

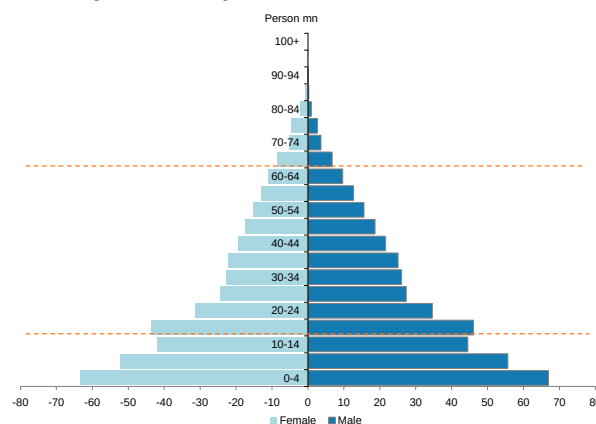
Declining Birth Rate



Sources: CEIC, NBS, Morgan Stanley Research.

Exhibit 17

China Population Pyramid: 1970



Sources: UN World Population Prospects (2008 Revision), Morgan Stanley Research.

Later, Longer, and Fewer

In 1971, the then-Premier Zhou Enlai proposed the first government campaign on population planning, aiming to reduce population growth in urban and rural areas to below 1% and 1.5%, respectively. Officially announced in December 1973, the policy called for "later, longer, and fewer"—short for "later marriages (and thus births), longer breaks between births, and fewer births." As a result, the birth rate declined from 33.4 births per 1,000 people in 1970 to just 18.2, while the death rate remained at a relatively low level. Nevertheless, due to the already high base of about 830 million in 1970, the total population rose to almost 1 billion by 1980, prompting the implementation of a more stringent population control policy.

⁴ "Working-age population" is defined as the number of people between the ages of 15 and 64.

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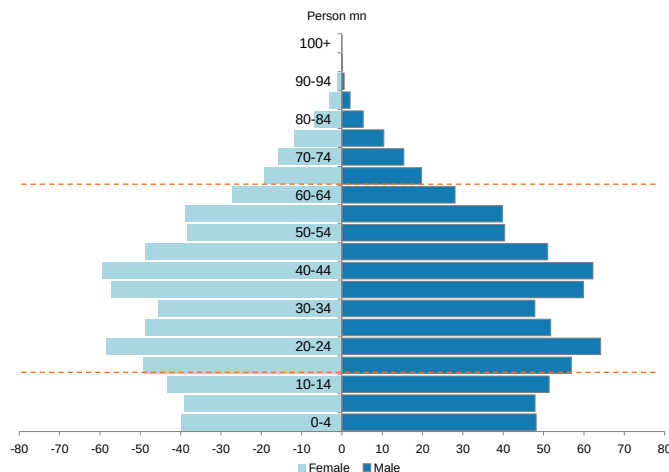
The One-Child Policy

The family planning policy, more commonly known as the “one-child policy,” was subsequently introduced in 1979. It allowed urban married couples to have only one child, with a few exceptions applied to rural couples, ethnic minorities, and couples where both individuals come from single-child families. Apparently, the policy is effective: The population growth rate dropped from 1.2% in 1978 to 0.4% in 2009.

According to UN data, total fertility in China dropped from about 2.6 children per woman in 1980 to about 2.0 in 1990, slightly below the replacement rate of 2 to 2.3. However, the combination of fewer births and a low mortality rate is leading to an increasingly “top-heavy” population pyramid (exhibits 18 and 19).

Exhibit 18

China Population Pyramid: 2010

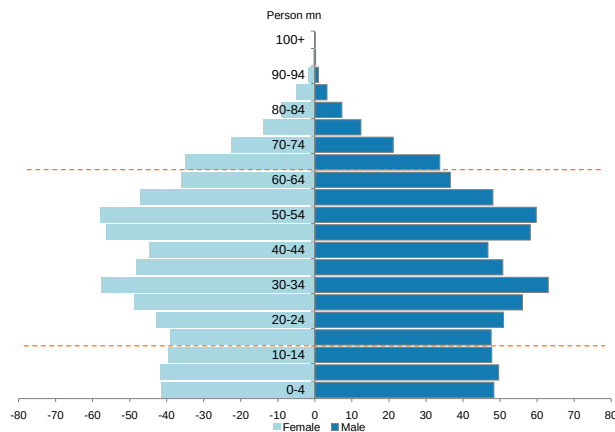


Sources: UN World Population Prospects (2008 Revision), Morgan Stanley Research.

The “heaviness” will shift upwards along the pyramid as the birth rate remains low and life expectancy continues to rise. Eventually, the total dependency ratio, computed by dividing the total number of child (aged 0 to 14) and elderly (aged 64 and above) dependents with the number of working-age population (aged 15 to 64), will bottom out, implying an increasing number of consuming dependents for each producing member of the population. According to UN estimates, China’s total dependency ratio will reach its trough and begin to rise some time within the next five years (exhibit 1).

Exhibit 19

China Population Pyramid: 2020



Source: UN World Population Prospects (2008 Revision), Morgan Stanley Research

The Fear: Diminishing Demographic Dividend

The term “demographic dividend” has been used extensively to refer to the favorable effects on economic growth of having a high percentage of the population being productive. It falls into place as a country evolves from a primarily rural society with high fertility and mortality rates, to a more developed urban society with low fertility and mortality rates. While in the end the transition will inevitably lead to an aging population, in the interim, the number of child dependents declines as the fertility rate decreases.

Before the population begins to age, the combination of fewer child dependents, a bigger working-age population, and a relatively small elderly population will create a demographic “golden age,” characterized by an abundant labor supply and low dependency ratio that is conducive to strong economic and per-capita income growth.

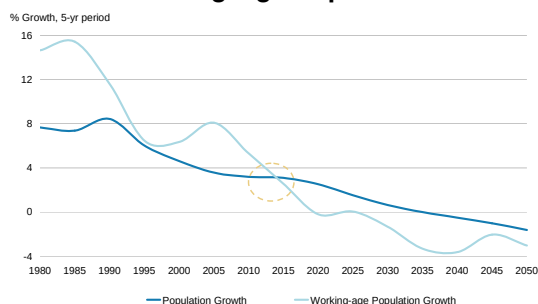
Eventually, the decreasing fertility rate reduces labor supply growth, and the lower mortality rate will lead to higher life expectancy and accelerate the growth of the elderly as a percentage of the population. The resulting rise in old-age dependency and the shrinking labor supply become a drag on further per-capita income growth.

This temporary “bonus” accompanies a demographic transition and drives economic growth through the abundant labor supply and the favorable mix between the producing and the consuming population. This bonus is, however, by definition, transitory in nature and will be depleted as total population grows faster than the working age population.

According to official demographic projections, this is exactly what will happen in China in the coming decade, with growth in the working-age population decelerating to a pace below that of the total population (exhibit 20).

Exhibit 20

Slowdown in Working-Age Population Growth



Sources: UN World Population Prospects (2008 Revision), Morgan Stanley Research.

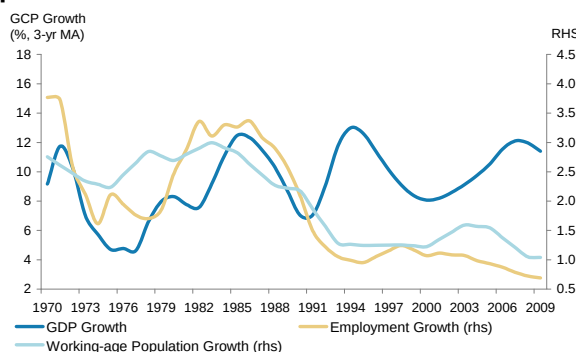
The Reality: Distinct Disconnect between Working-Age Population/Employment and Economic Growth

While the strong growth in the working-age population contributes to economic growth directly through the increase in labor supply, it is only part of the story. Theoretically, the working-age population can be considered the sum of the “economically active” and “non-economically active” population, with the former further broken out into the “employed” and “unemployed.” It is the “employed” within the working-age population, who are actively engaged in producing activities, which directly contribute to economic growth.

While the “quantity” dimension certainly matters in the context of a strong working-age population and thus a large number of employed workers, it is the “quality” factors boosting labor productivity that facilitate the feeding through and that have the maximum impact on economic growth.

Exhibit 21

Distinct Disconnect between GDP and Working-Age Population Growth in China



Sources: CEIC UN World Population Prospects (2008 Revision), Morgan Stanley Research.

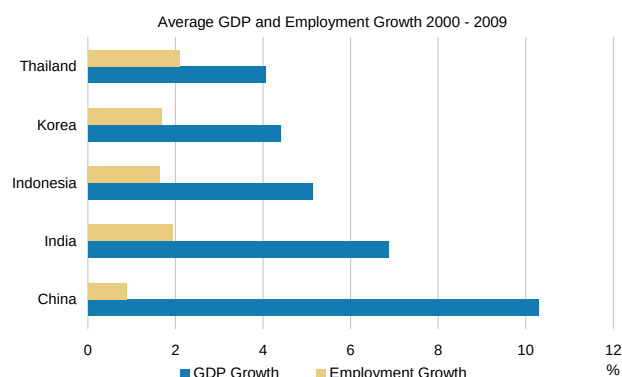
China has illustrated this over the past two decades: There is a sizable gap between GDP growth rate and the growth rates of the working-age population and employment (exhibit 21). In particular, while there was significant correlation between GDP and employment growth and—to a lesser extent—working wage population growth during 1970s and 1980s in China, this correlation has broken down completely since early 1990s

- Average GDP growth during 1970-90 was 8.4%, but average working-age population and employment growth was 2.58% and 2.55%, respectively. The gap is about 5-6 percentage points.
- Average GDP growth rate per annum between 1990 and 2010 was 10.5%, while average growth for the working-age population and employment was merely 1.3% and 0.98%, respectively. The gap is 9-10 percentage points.
- GDP growth accelerated to 14% in 1993 from 5.7% in 1990, while employment growth slowed to 1.0% from 2.1%.
- GDP growth accelerated to 11% in 2008 from 8.1% in 2000, while employment growth slowed to 0.7% from 1.1%.

This distinct disconnect between economic growth and employment growth in China is further highlighted by cross-country comparison. China has managed to deliver the most rapid growth (10.3% per annum on average) over the past decade of key Asian countries—India, Indonesia, Korea, and Thailand; however, China’s extraordinarily strong growth has entailed the least support from employment growth compared to its Asian peers (exhibit 22).

Exhibit 22

China’s Jobless Growth



Sources: CEIC, Morgan Stanley Research.

China’s strong growth over the past two decades clearly reflects rapid labor productivity growth, while the quantity of labor supply has played a rather insignificant role. The impressive labor productivity gains were catalyzed by the

profound structural reform carried out by the Chinese authorities since the early 1980s. Specifically, greater labor productivity can be attributed to:

- The reallocation of labor from low- to high-productivity activities (from farming in rural areas to urban manufacturing).
- Human capital investment—productivity gains stemming from higher quality labor as a result of education, training, learning by doing, and so on.
- Capital deepening—productivity gains stemming from a higher capital-labor ratio as a result of persistently strong investment growth, underpinned by a high savings ratio.
- Technological innovation and structural reform that contribute to organic improvement in total factor productivity.

The key question is whether there is room for further substantial labor productivity gains in China or whether the inevitable slowdown in working-age population growth will prevent strong growth from lasting over the next decade.

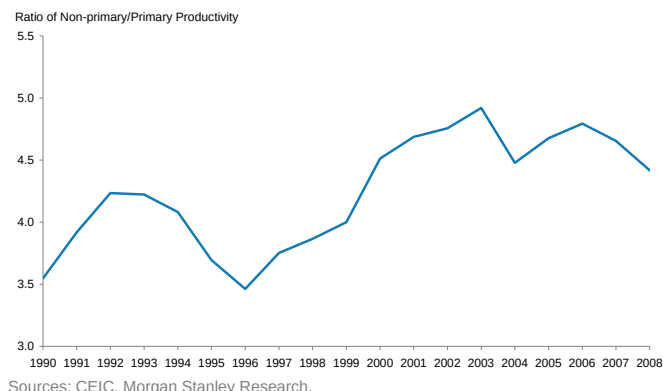
Significant Potential for Sustained Productivity Gains Stemming from Labor Reallocation

A popular current view among some observers is that, after three decades of economic reform that initially started in rural areas, there is little scope for the “low-hanging-fruit” type of labor productivity gains stemming from transferring labor from low- to high-productivity activity. We disagree.

- There is a wide gap in labor productivity between primary and non-primary industries in China, pointing to significant potential for productivity gains through the transfer of labor between industries. Specifically, labor in non-primary industries (e.g., manufacturing and service) has consistently been 4.5-5.0 times as productive as labor in primary industry (e.g., farming and other agricultural activities) since the early 2000s (exhibit 23). The gap in labor productivity between non-primary and primary industries has in fact widened substantially in 2000-09 relative to the previous decade (1990-99).
- Primary industry still has a disproportionately large share of total employment in China, suggesting an overhang of surplus labor in primary industry in rural areas. The primary sector represented as much as 35% of total employment in 2008. Although this ratio has already fallen significantly—from 45% 10 years ago and 55% 20 years ago—it is still 1.7 and 2.0 times as high as the shares of Korea and Japan, respectively, when the latter two were at a stage of development similar to that of China at present (Exhibit 24). (See *Not Whether but How Growth Will Decelerate* on page 4 of this report.)

Exhibit 23

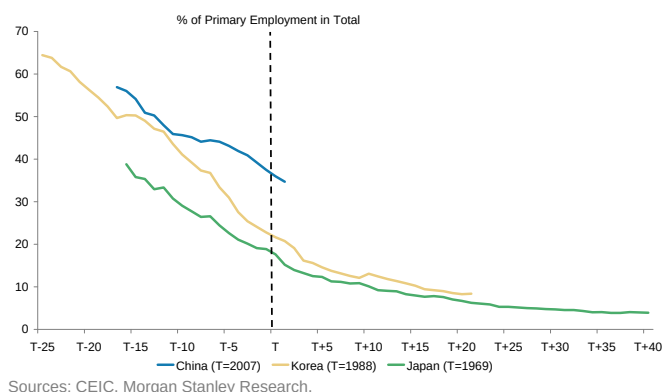
Large Labor Productivity Gap between Non-primary and Primary Industries



- Looking ahead, we see enormous potential for labor productivity gains in the primary sector in China, as was the case for both Japan and Korea. Labor productivity in the primary sector in both Japan and Korea increased by about 50% during the 10 years after their economies crossed the inflection point, as described in the earlier section (exhibit 25). The pace of labor productivity growth in China is clearly faster than in either Japan or Korea around the inflection point (Year T), as indicated by the steeper slope of productivity index for China in exhibit 11. If this pace were to be maintained through the next decade, the share of primary sector employment would decline rapidly, continuing to make a substantial amount of labor available for high-productivity activity.

Exhibit 24

Disproportionately Large Share of Primary Industry Employment in China



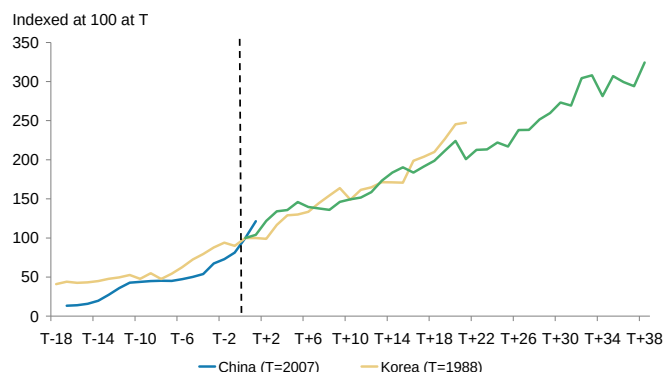
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Exhibit 25

Enormous Upside for Labor Productivity Gain in Primary Industry in China



Sources: CEIC, Morgan Stanley Research.

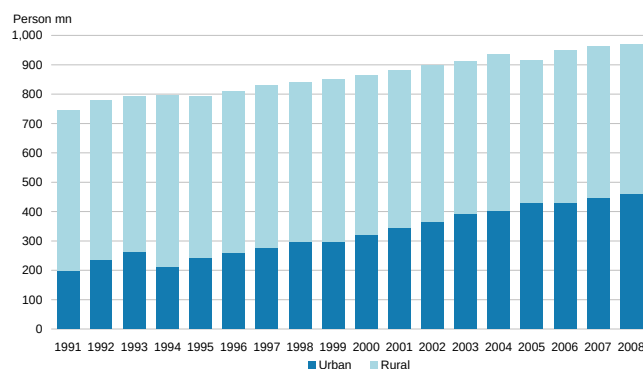
Rural Surplus Labor: How Much Exactly, If Any?

Despite having one of the largest working-age populations in the world, anecdotal evidence suggests that China's labor market is becoming increasingly tight. Some contend that this is evidence of the imminent exhaustion of the rural surplus labor supply, not to mention that the rural community is perceived to be aging a lot more rapidly than its urban counterpart.

However, a closer examination of the data suggests a less gloomy outlook. According to official statistics, as of 2008, the rural working-age population was 510 million strong, compared to 462 million in urban areas (exhibit 26). And while the rural population is aging just a tad faster than the rest of the country, but the working-age population still accounted for more than 70% of the total (exhibit 27).

Exhibit 26

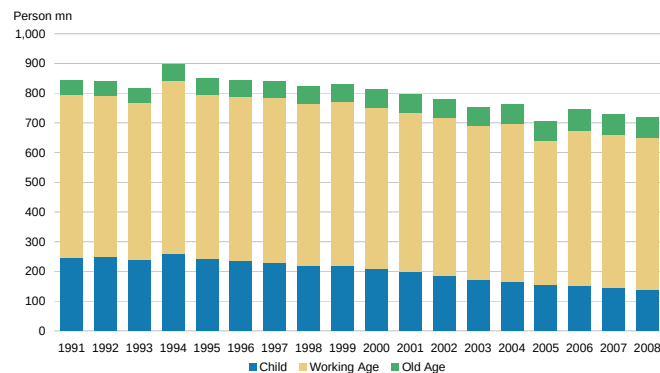
Rural Working-Age Population Is Huge...Despite Shrinking Gradually



Sources: CEIC, Morgan Stanley Research.

Exhibit 27

Rural Population—Age Structure



Sources: CEIC, NBS, Morgan Stanley Research.

We estimate there could be 80 to 100 million workers in rural areas that could transfer to non-primary industries and/or reallocate to urban areas. Specifically:

- Based on the analysis of the agricultural output and productivity levels, we believe that about 197 million workers were needed to produce 2008's output.⁵ It follows that there were about 313 million rural workers in excess of what was needed to support the level of agricultural activities in 2008. This is the part of the working-age population that can either be engaged in rural non-farm activities or is ready to relocate in pursuit of job opportunities elsewhere.
- As of 2008, there were about 124 million migrant workers in urban areas. While this includes migrations from city to city—a huge number, if we assume that all migrant workers are from rural areas—this leaves 189 million workers who could take up non-farm jobs locally or in urban areas.
- In terms of age structure, most surveys suggest that it is the older workers in the labor force who have stayed in the villages. It is reasonable to assume that the 140 million workers aged 50 to 64 would be part of the 197 million workers needed to support agricultural production.
- If we further assume that the remaining 57 million workers are aged 30-49, this would mean 313 million workers are ready to take up non-farm jobs (145 million workers aged 15-29 and 168 million aged 30-49). Again, assuming that all migrant workers are aged 15-29, and only half of those aged between 30 and 49 are willing to relocate, it seems safe to conclude that there were 80-100 million surplus rural workers who could potentially migrate to non-farm jobs in urban areas (exhibit 28).

⁵ The analysis is based on data from the NBS, Ministry of Agriculture (MoA) and information from Dr. Zhong Yu and Lan Haitao's article, "Analysis on Rural Surplus Labor: State and Trend", 2009

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Exhibit 28

Breakdown of Rural Working-Age Population

Age		Job Nature / Status	
15-29	145 mn	Migrant	124 mn
30-49	168 mn	Non-farm, locally or elsewhere	189 mn
	57 mn	Rural Farm (Required)	197 mn
50-64	140 mn		

Sources: CEIC, China Statistical Yearbook 2008, Morgan Stanley Research.

Policies to Boost Labor Supply by Facilitating the Efficiency-Enhancing Transfer of Labor

There is definitely substantial room for more efficient allocation of labor in China: Even though only 197 million workers were required to support the agricultural production in 2008, there were 270 million rural workers engaged in farming activities. To this end, it is important to make sure that the proper incentives and necessary infrastructure—both hard (like transportation) and soft (like public services)—are provided to rural surplus labor to allow them to migrate out of the villages and find jobs in the urban area.

Crisis leads to reform, and China is no exception. While market-oriented reform, opening-up, and urbanization (MOU) are the three pillars of China's remarkable economic success over the past three decades, progress on urbanization has thus far been largely a result of passive adaptation to a rapidly changing reality that is being shaped by market-oriented reform and opening-up. The recent recession has made it more important for China to promote domestic demand with a view to bringing about sustainable growth in the long run. Against this background, urbanization—where reform initiatives have been lagging—is being given a greater role in helping to deliver continued strong growth. To this end, Chinese authorities have recently begun to call for an acceleration of urbanization by removing some of the barriers to rural-urban migration (see *China Economics: One Country, Three Economies: Urbanization as a Primary Driver of Growth*, March 3, 2010).

Of particular note, three important policy initiatives are already in progress under the broad umbrella of facilitating urbanization:

- The “Go West” development program
- Deregulation of household registration system
- The ambitious social housing program

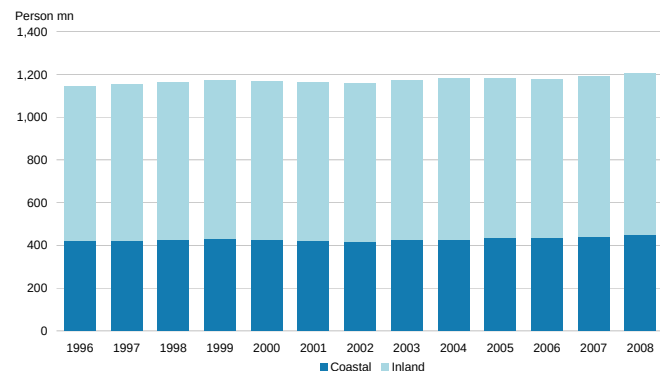
First, the Chinese government's massive “Go West” development program provides additional impetus to the more efficient allocation of labor resources. This program provides the necessary hard infrastructure to facilitate rural-urban, inland-coastal labor migration. China is unusual in that there is considerable disparity between the eastern (or coastal) region and the central and western (or inland) regions in terms of economic development, industry mix, and demographic depth. Of the registered population (which excludes migrant workers), the inland region outnumbers the coastal region almost two to one (exhibit 29). For instance, the southwestern Sichuan Province alone had a registered population of almost 80 million in 2008, equal to the total population of Germany.

The demographic depth of the inland region is well suited for labor-intensive manufacturing production, and manufacturers know this: Foreign companies have been aggressively moving inland. Average foreign direct investment (FDI) year over year growth in the past decade reached almost 20% for inland region, compared with about 11% in coastal region (exhibit 30), and inland fixed asset investments have quadrupled in the past five years (exhibit 31). According to a recent study by the American Chamber of Commerce, more than half of the 270 companies surveyed in 2009 had already expanded their businesses beyond the coastal region.

Chongqing, in particular, has had success in attracting foreign investment: Hewlett-Packard (HP), the largest laptop manufacturer in the world, has begun producing computers from its manufacturing base there. This move will help boost HP's production capacity to 60 million units annually in 2012, with 30 million units manufactured and assembled in Chongqing.

Exhibit 29

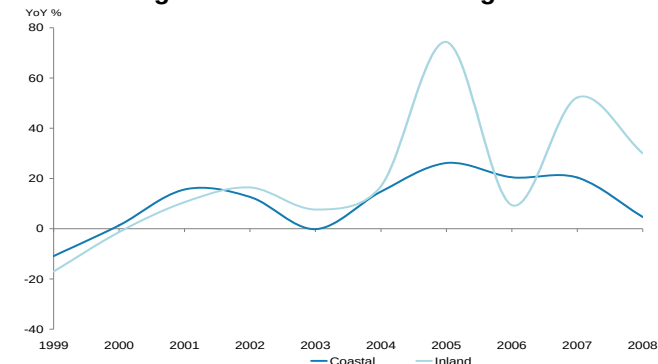
Inland "Outnumbered" Coastal Region



Besides HP, a large number of original design manufacturers (ODM), including Foxconn, Quanta, Inventec, and Compal, plan to follow HP's lead into Chongqing and in sum will churn out more than 80 million units annually, compared to the 160 million units produced worldwide in 2008, of which about 100 million units were produced in China. More than 80 parts-makers also have a presence in the city to provide support to the big names. This will transform Chongqing into the largest laptop manufacturing base in China and, indeed, Asia.

Exhibit 30

Accelerating FDI Growth in Inland Region



Beyond Chongqing, Intel built an assembly plant in Chengdu, the capital city of Sichuan province, in 2005, and Dell decided to set up its second operation center there. Dell is expected to invest US\$10 billion in China over the next 10 years.

Exhibit 31

Inland FAI Quadrupled from 2004 to 2009

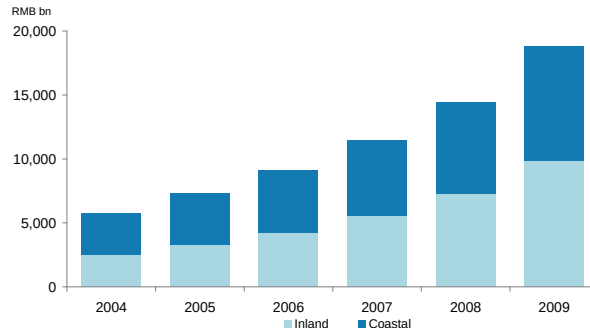
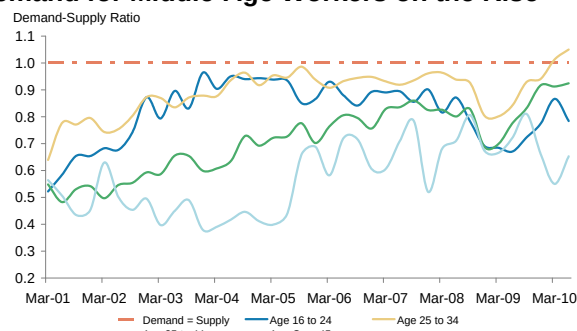


Exhibit 32

Demand for Middle-Age Workers on the Rise



Obviously, the labor market table has turned. Before it was workers travelling thousands of miles in search of better jobs, now it is the other way round. Manufacturers are also becoming less selective; the demand for middle-age workers has been on the rise since year 2001 (exhibit 32). Workers from older age groups are now able to contribute more actively to the production activities as health conditions improve. Since 2000, medical and healthcare expenditure has surged from less than Rmb 1 billion to more than Rmb 127 billion, an increase of more than 170 times.

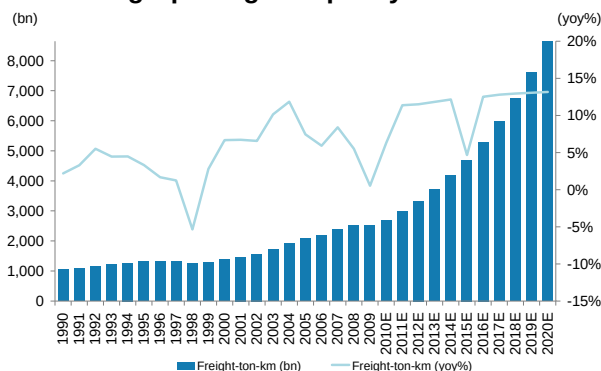
On a side note, we believe that worries that inland labor migration will suffer because of great travel distances and potential logistic gridlocks are overstated.

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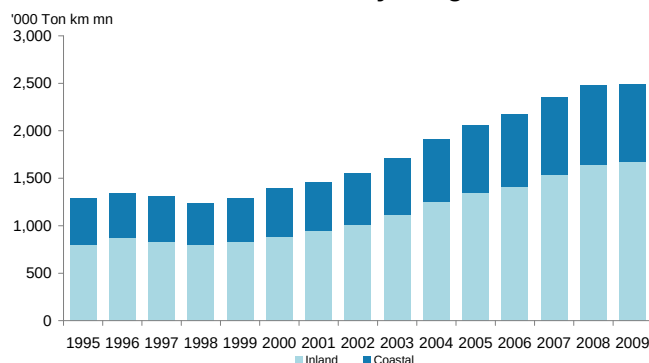
Exhibit 33

HSR Freeing Up Freight Capacity

Sources: CEIC, MoR, Morgan Stanley Research.

According to a detailed analysis from our Transportation and Construction & Infrastructure teams (see *China Railroads: Capitalizing on Opportunities from Potential Industry Reform*, January 7, 2010 and *China Railroads, HSR Gives Boost to Freight Capacity*; OW DQR & GSR, August 13, 2010), China's Ministry of Railways estimated that the existing high-speed rail services, which make up only 53% of the 13,000-kilometer target set for 2012, should provide additional annual freight capacity of over 230 million tons for rail transportation, accounting for about 7% of 3.3 billion tons of rail freight demand in 2009. With China's rail track length due to total 183,796 km by 2020, freight capacity is set to increase for the next 10 years (exhibit 33). Historically, inland railway freight carried has been growing faster than coastal freight (exhibit 34). If the trend continues, as we believe it will, the additional capacity should help to accommodate the increased freight traffic.

Exhibit 34

Solid Growth in Inland Railway Freight Carried

Sources: CEIC, Morgan Stanley Research.

Second, the deregulation of the household registration system that has restricted rural-urban labor mobility for over half a century in China is making encouraging progress, providing the necessary soft infrastructure to facilitate efficiency-enhancing transfer of labor.

Indeed, one of the major obstacles for migrant workers seeking to settle in the cities is the *de facto* discrimination in terms of employment, social security, public services, education, and healthcare coverage compared to urban residents. These all come with the urban *hukuo*, or household registration system, and are not available for rural households. The lack of institutions to facilitate the smooth assimilation of rural workers into urban life has been one of the biggest challenges to ensuring a stable workforce in the cities.

Since the beginning of 2010, Chinese authorities have been calling for reform of the household registration system with a view to accelerating urbanization. The relevant deregulation will initially be applied to small- and medium-sized cities and to migrant workers who already have a job in the urban area and their family members and relatives.

Ambitious urbanization programs are also being implemented in selected mid- and large-sized cities under pilot programs. For example, Shijiazhuang, the capital city of Hebei province, allows migrant workers to register as urban householders in the city as long as they sign labor contracts of two years or longer. Chongqing, one of the four cities and the largest to report directly to the central government, is implementing an ambitious and comprehensive program to accelerate urbanization on a pilot basis (see box). It is expected that the Chongqing model of urbanization, if successful, will be rolled out to the rest of country over the coming decade.

Household Registration Reform: The Case of Chongqing

Chongqing's household registration reform officially began in July 2010. Its scale is unprecedented in modern Chinese history, and its target is extremely aggressive. A pilot city for rural-urban integration since 2007, its progress will be closely watched by policymakers and market observers. The outcome will inevitably have important implications for similar reforms in other parts of the country over the coming decade.

An aggressive target: To migrate and offer urban household status to 10 million of rural population by year 2020, pushing the urbanization rate from 29% to 60%.

Potential Headwinds

- **Financing.** The migration of rural household involves putting on five "urban dresses" (employment, housing, education, medical, and social security) and taking off three "rural sweaters" (the land contractual management rights, homestead rights, and forest property rights). It is estimated to cost about Rmb40,000 per head for the former, and another Rmb70 billion for the latter in the first 3 years. Total costs could add up to more than Rmb600 billion over the next 10 years.
- **Bottleneck in urban land supply.** According to Chongqing's Ministry of Land Resources and Housing Management, since 2007 Chongqing's land demand stemming from urbanization outstrips the land supply determined by the central government by 50 square kilometers.

The Roadmap Ahead

- **The two-step approach.** During the first stage (now to year 2011), registration will cover more than 3 million rural workers, including 2.3 million migrant workers who had worked for more than three years in towns and five years in cities, 0.7 million college students with rural registration, and 0.4 million people with their "rural sweaters" already taken off. The second stage (2012 to 2020) will involve the systematic conversions of the remaining 7 million rural residents into urban residents.
- **Financing from public and private sources.** Medical and social security entitlements for incoming rural workers, which add up to 80% of the cost for the five "urban dresses," will be provided by the employers in compliance with related regulations. Investments in education will come from the government budget, and 40 million square meters of public rental housing for rural migrant workers will be built over the next 10 years, costing about Rmb100 billion.
- **The cost associated with "taking off the three rural sweaters,"** that is, compensation for surrendering rural land rights, will be transferred to the private sector via "land contract" trading on the rural land exchange. It allows rural residents to monetize on their land ownership rights and offers an alternative source of land supply.

The implication for a successful implementation is profound. The successful implementation of this comprehensive household registration reform will not only encourage the migration of young rural workers to the cities, but it will also help improve the dependency ratio in the urban areas and in turn alleviate the pressure on public finances from the pension system. Last but not least, the increase in the urban population and the upgrade in social welfare for the newcomers will be a powerful driver to promote consumption.

Third, the aggressive social housing program, the progress of which we have high conviction on, is also among one of the most important policies. (See *China Economics: Social Housing: Lackluster Growth or Quantum Jump?*, August 13, 2010.) Housing costs have always been a big-outlay item, especially for migrant workers. Historically, property developers had little incentive to build economic or public rental housing to accommodate the needs of the low-income group.

In the end, many of the migrant workers retreat to outlying areas in the cities where living conditions are poor and there is little access to basic infrastructure. This recurring issue has now evolved into a gridlock preventing migrant workers from assimilating into the cities. The social housing program, which provides affordable housing at a reasonable rent and in convenient locations, comes as a powerful resolution to the issue. The better living conditions will improve the quality of life for migrant workers and discourage frequent job changes, which will in turn increase labor hours worked and hence labor supply.

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Forecast of Labor Supply through 2020

We forecast that labor supply will remain abundant in China through 2020:

- The number in urban and rural non-farm employment will increase by 80 million between 2010 and 2020, even though the working-age population is expected to increase by only 20 million (exhibit 35).
- The ratio of urban and non-farm employment in total will increase from 67% in 2010 to 75% in 2020.
- While the average rate of growth in the working-age population will be only 0.22% per annum, and urban and non-farm employment will increase at a rate of 1.4% per annum on average (exhibit 36).

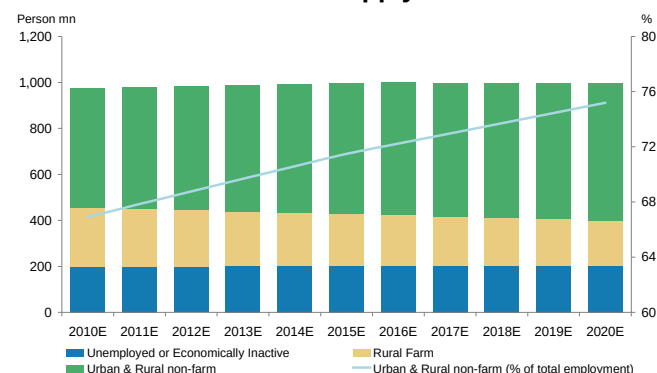
Our forecasts are based on the following technical assumptions:

- The labor force participation rate (total employment/working-age population) is unchanged through 2020;
- Agricultural production and productivity increase at about the same rate so that the size of the agricultural workforce is unchanged. This is a conservative assumption given the low level of mechanization prevalent in rural China; and
- Rural farm employment gradually approaches the level commensurate with underlying labor productivity by 2020.

We expect that the increase in urban and rural non-farm employment due to reallocation of labor would take place against the backdrop of continued rapid urbanization, aided by an enabling policy environment over the next decade.

Exhibit 35

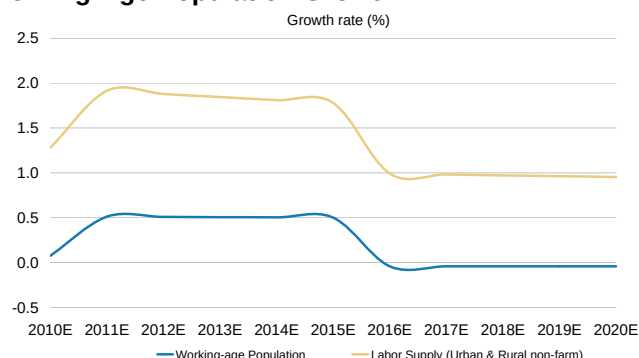
Potential Boost in Labor Supply



Sources: UN World Population Prospects (2008 Revision), CEIC, Morgan Stanley Research.

Exhibit 36

Labor Supply Growth to Continue to Outpace Working-Age Population Growth



Sources: UN World Population Prospects (2008 Revision), CEIC, Morgan Stanley Research.

Risks to Forecasts and Other Potential Factors Contributing to Labor Productivity Improvement

To the upside, the technical assumptions we made are rather conservative and the actual outcome could easily turn out to be better. On the downside, policy implementation to facilitate urbanization and labor mobility may not follow through as we envisage. That said, policy implementation is likely to be a decreasing function of the actual labor supply: If labor supply growth is robust without policy support, the urgency for proactive implementation of the relevant policies will likely decrease.

Besides reallocation of labor from low-productivity to high-productivity activities, strengthened human capital investment and continued capital deepening—the other two key sources of labor productivity gains—are likely to persist in China, making the growth of working-age population unlikely a binding constraint for overall economic growth through 2020, in our view.

- On human capital investment, according to a study done by the authoritative demographics economist Dr. Cai Fang, an extra year of education would lead to 17% increase in labor productivity in the manufacturing industry.⁶ And the labor productivity for a manufacturing company could increase by 66% if those who now have below-junior-high-school qualifications were to continue to undergraduate education. In this regard, China stands to benefit from the very strong growth in human capital investment since early 2000: While secondary or higher school enrollment increased to and stabilized at around 40%, the college student enrollment surged from 2.1% of the total in 1999 to 11.3% in 2009 (exhibit 37).

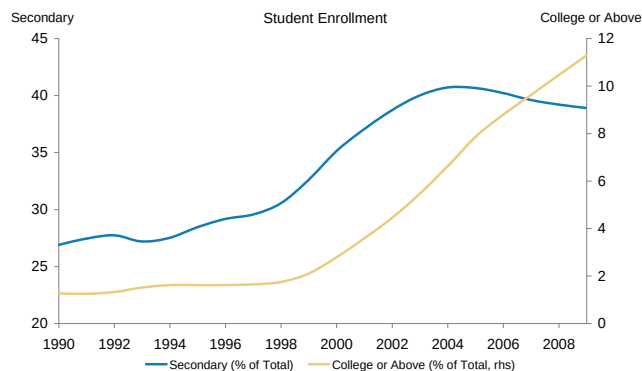
⁶ For further details please refer to Dr. Cai Fang's article, "Demographic Dividend: Continuity or Substitution," 2009.

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Exhibit 37

China: Massive Investment in Human Capital

- Physical capital deepening will continue in China through 2020, although possibly not as fast as in the past decade, in our view. The resultant higher capital-labor ratio will contribute to sustained labor productivity gain in China. We will discuss the impact of “second demographic dividend”—

namely, the renewed incentive to save and thus invest in preparation for the extended period of retirement on the back of rising life expectancy—in a later report.

Last But Not Least, a Caveat

The sixth National Population Census will be carried out in November of this year. The data compilation and analyses should be done by December 2011. We expect this to confirm speculation that China's population is under-reported, especially in the birth rate and for the younger part of the population, in part due to evasion of the one-child policy. Consequently, we expect the report to paint a more favorable picture than the now widely cited UN estimation, which we believe overstates the pace of demographic transition in China. It will likely turn out to be a pleasant surprise for us China observers and serve to strengthen our view that labor supply will remain abundant and demographics will unlikely be headwinds to economic growth through 2020.

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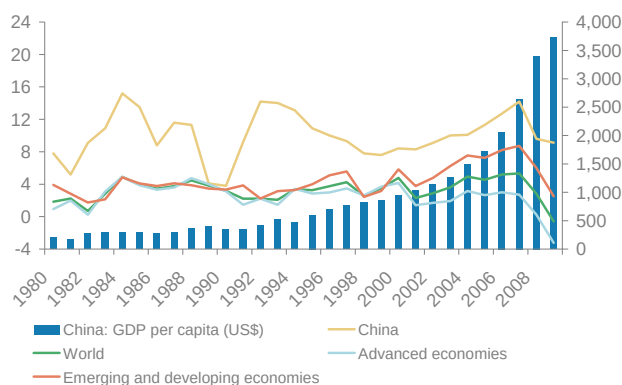
A Golden Age for Consumption

China's economic achievements since the launch of economic reform in 1978 have been extraordinary. By 2009, China's nominal per-capita GDP had reached US\$3,679, or 16 times 1978's level of US\$226, representing a real GDP CAGR of 9.5% across the period, outperforming not only the developed economies but also developing peers by a wide margin (exhibit 37).

Investment and exports have been the primary drivers of China's strong growth, while consumption growth has underperformed: Its share of GDP declined by nearly 14 percentage points over from 2000-09 (exhibit 38).

Exhibit 37

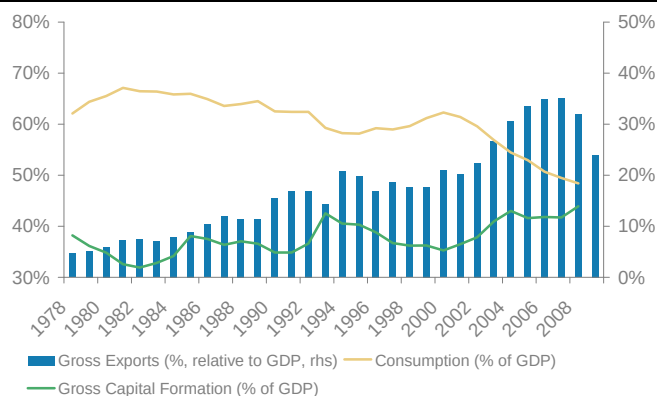
China: Outperformance in Overall Economic Growth



Sources: IMF, Morgan Stanley Research.

Exhibit 38

China: Consumption Underperformed as a Growth Driver



Sources: CEIC, Morgan Stanley Research.

As a result, by 2009, China's consumption-GDP ratio was significantly below not only those of high-income countries,

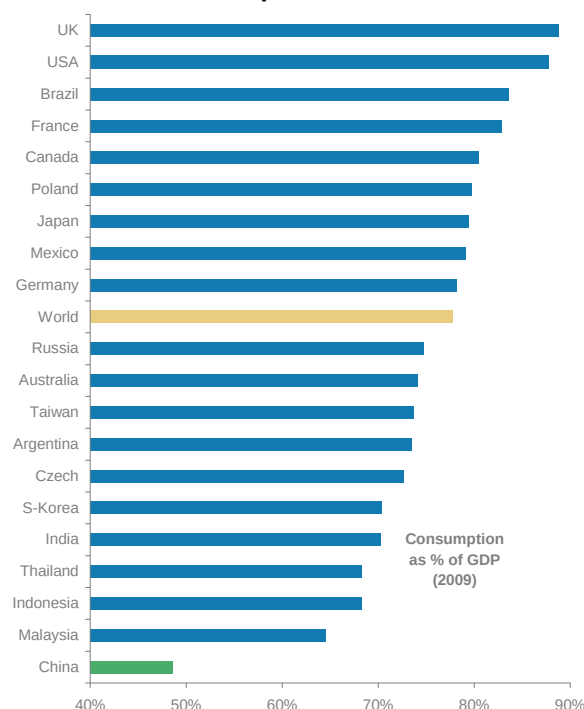
like the US, and middle-income peers, like Malaysia, but also those of low-income countries, like India (exhibit 39).

These comparisons have helped form a consensus among most China observers that there is serious under-consumption in China and that a substantial boost to consumption is required to ensure more sustainable and balanced growth. On this subject, some China observers have become much more concerned as they fear that this is not only an issue of rebalancing China's economy over the long run but also of economic stability in the short run. Some China bears even predict that the Chinese economy is about to implode, as the consumption-GDP ratio in China is simply so unusually high, and thus fragile, that economic growth could easily collapse in face of a major shock.

While we share the consensus view that China's consumption is relatively weak, we dismiss the rather alarmist view that the Chinese economy is so seriously imbalanced as to pose a threat to economic stability in the short run. This is because we believe China's official statistics substantially understate the true magnitude of consumption (especially the consumption of services) in China. We address this topic further later in this report.

Exhibit 39

China's Under-consumption



Sources: Euromonitor, CEIC, Morgan Stanley Research.

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China's Consumption at an Inflection Point: Empirical Evidence

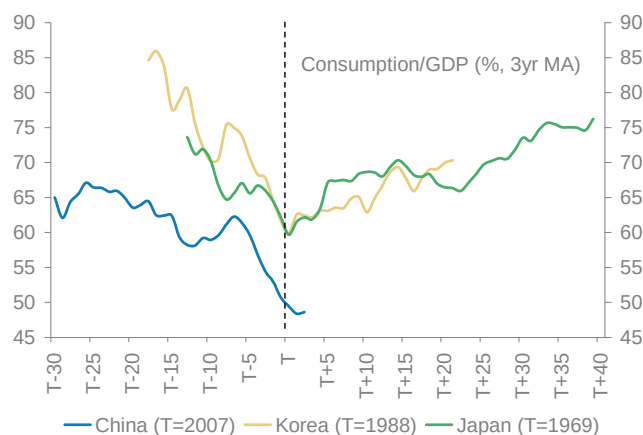
China's economy is at an inflection point. Beyond this point, over the next decade, we believe that consumption is likely to outperform quite strongly.

As argued in the first section of this report, the Chinese economy is at an inflection point similar to that of Japan in 1969 and of Korea in 1988. History would suggest that beyond this inflection point the economic structure tends to undergo profound transformation, with the three key ratios of the economy—consumption-GDP, service sector-GDP, and labor income-GDP—rising rapidly.

For instance, Japan's consumption-GDP ratio increased from 60% in 1969 to 69% in 1979, and Korea's from 60% in 1988 to 65% in 1998, as consumption growth began to significantly outpace overall economic growth.

Exhibit 40

China's Consumption at an Inflection Point: Historical Precedents



Source: Morgan Stanley Research.

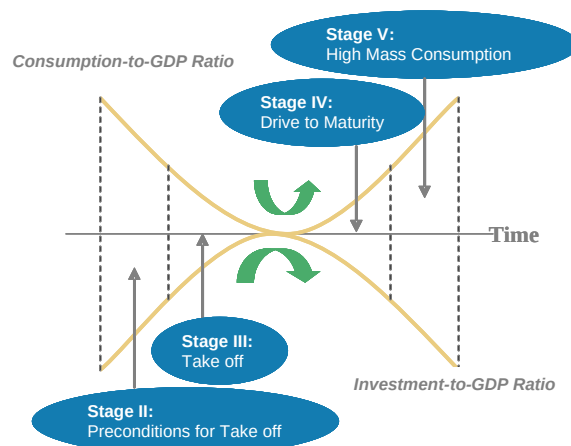
China's Consumption at an Inflection Point: Theory

In our view, a country is justified in accumulating physical capital and wealth at the early stage of development in order to pave the way for the transition towards more consumption-driven growth. For a developing economy, savings is an essential and integral part of industrialization process. History has shown that the only way to industrialize an economy is to increase the capital-labor ratio so that poor farmers can be equipped with industrial machines and equipment to produce goods with a higher value than that of farm goods (exhibit 40). To install that piece of machinery, you need to save and invest.

There are a number of theories that shed light on the development stages of a country, with the most popular ones including the U Curve Theory, Rostow's Stages of Growth, and Chenery's Division of Industrialization Stages.

Exhibit 41

China's Consumption at an Inflection Point: An Analytical Framework



Source: Morgan Stanley Research.

- The U Curve Theory is the most straightforward explanation for the trajectory of economic transition in which the investment intensity of the economy (or investment-GDP ratio) keeps rising during the early state of industrialization until an inflection point is reached. Beyond that point, the consumption intensity of the economy (consumption-GDP ratio) starts to bottom out.
- Rostow's Stages of Growth is one of the major analytical frameworks that explain the pattern of economic development. It postulates that economic modernization occurs in five basic stages of varying length, featuring traditional society, preconditions for take-off, take-off, drive to maturity, and age of high mass consumption. Rostow asserts that countries go through each of these stages fairly linearly and describes a number of conditions that would likely occur to investment, consumption, and social trends at each stage.⁷

We unify the two theories—the U Curve and Rostow's Stages of Growth—into one framework to help illustrate the potential consumption trends in China (Exhibit 41). According to Rostow's Stages of Growth theory, an economy's development divides into five stages. China appears to be passing stage III of "takeoff" and poised to transition into the stage IV of "drive to maturity." The "drive to maturity" features

⁷ W. W. Rostow. *The Stages of Economic Growth: A Non-Communist Manifesto* Cambridge University Press (1960).

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a rebalancing among sectors, great poverty reduction, and an improved standard of living as the society no longer needs to sacrifice its comfort in order to strengthen certain sectors. Fitting the Rostow's Stages of Growth into the U Curve Theory framework, consumption intensity declines in the stages of "pre-conditions for takeoff" and "takeoff" but is set to rebound in the stage of "drive to maturity."

China's Consumption through 2020: Three Illustrative Scenarios

We construct three scenarios to help illustrate how consumption in China will likely evolve through 2020 relative to that of US and the world economies. These analyses are based on the forecasts we laid out in the first section, which delineates the steps of the economic transition that developed economies such as Japan and Korea have undergone.

Exhibit 42

Assumptions for Forecasts under Different Scenarios

	Real GDP Growth (%)	CPI Inflation Growth (%)	Incremental Consumption Intensity (ppt)	USD/CNY Rate (end 2020)
	(average change per annum)			
Base-case	8.0	3.5	0.7	5.5
Alternative I	9.5	2.5	-0.5	6.0
Alternative II	6.5	4.0	1.4	5.0

Source: Morgan Stanley Research.

The key parameters under different scenarios are summarized in Exhibit 42:

- **Base case (70% probability).** We expect annual real GDP growth and CPI inflation to hit 8.0% and 3.5% per year in our base case. Meanwhile, using the experiences of Japan and Korea during the takeoff period of consumption as a benchmark, the increment of consumption intensity per year is set at 0.7 percentage points per annum, which would bring consumption intensity to 56% by 2020 from 49% in 2009. The US dollar/renminbi exchange rate is expected to reach 5.5 by 2020.
- **Alternative scenario I: Current trends continue (20% probability).** The Chinese economy continues doing business as usual, with no material change from the previous decade, featuring strong growth (GDP at 9.5% year over year), modest inflation (CPI at 2.5% year over year), and no meaningful transformation of the economic structure (consumption intensity slides 0.5 percentage points per annum to 44% in 2020). Renminbi exchange rate mechanism reform is slower than expected, so that the US dollar/renminbi exchange rate reaches 6.0 by 2020.
- **Alternative scenario II: A Japanese-style adjustment (10% probability).** A Japan-style transition featuring a drastic deceleration of growth (GDP at 6.5% year over

year) and structural adjustment (consumption intensity improves 1.4 percentage points per annum to 63% in 2020). Such a scenario could be catalyzed either by very proactive (perhaps draconian) policy intervention to artificially correct the structure of the economy or external shocks such as a complete meltdown in external demand and a sustained surge in international commodities prices due to supply shocks. The pace of renminbi foreign exchange reform accelerates under the second scenario (US dollar/renminbi exchange rate is at 5 by 2020).

Exhibit 43

Assumptions for Forecasts for US and World Economies

	Real GDP Growth (%)	CPI Inflation Growth (%)	Incremental Consumption Intensity (ppt)
	(average change per annum)		
US	2.7	1.6	-0.5
World	4.6	2.9	0.0

Sources: IMF, Morgan Stanley Research.

US/World Growth. In the aftermath of the global financial crisis, we assume trend growth in the US to be low, averaging 2.7% of real GDP growth and 1.6% of CPI inflation during 2010-20. This assumption is based on the IMF's latest *World Economic Outlook* for 2010-15, with a moving average of the previous three years for 2016-20. Meanwhile, reflecting the need for US households to repair their balance sheets by saving more, we assume that the consumption intensity in the US would decline 0.5 percentage points per annum. Finally, we assume trend growth of world GDP of 4.5%, with 2.9% of average CPI inflation during 2011-20 (IMF forecasts during 2010-15 and rolling average of previous three years for 2016-20), while the consumption intensity would stay unchanged at its current level (Exhibit 43).

The Results from Scenario Analyses

- China's incremental consumption in US dollar terms overtook that of the US for the first time in 2008 (exhibit 44). While incremental consumption for the US was negative in 2009 as the result of financial crisis, China still managed to maintain expansion, suggesting that it had replaced the US as a primary driver of global consumption growth. Going forward, we expect China's incremental consumption to dwarf that of the US, thanks to faster headline GDP growth and rising consumption intensity as the underlying economic structure evolves. By 2020, we estimate China's annual incremental consumption should be roughly double the US's under our base-case scenario (exhibit 8).

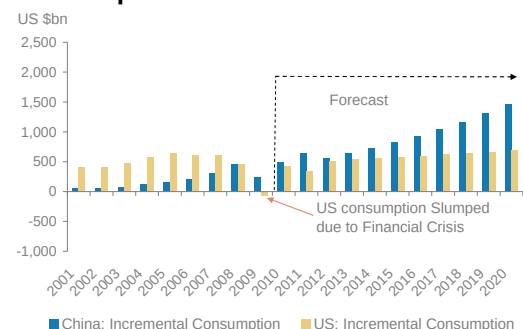
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Exhibit 44

China: Illustrative Forecasts of Incremental Consumption under Base-Case Scenario

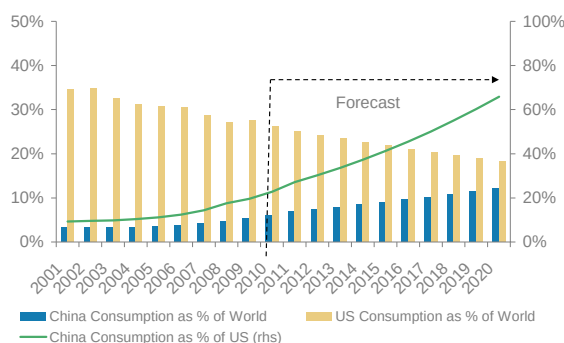


Sources: CEIC, Morgan Stanley Research.

- **Total consumption.** In our base-case scenario, China's consumption would reach two-thirds of the US level by 2020, from about 9% in 2000 and 20% in 2010. Meanwhile, the contribution of China to world total consumption will rise to 12% by 2020, from 3% in 2000 and 5.4% in 2010, while the US contribution will decline to 18.3% by 2020 from 27% in 2010 (exhibit 45).
- **Additional colors from alternative scenarios.** In terms of incremental consumption, China will reach 1.2 times US and 12% of world total under alternative scenario I, but 2.5 times of US and 24% of world total under alternative scenario II. In terms of total consumption, China will reach 49% of US and 9% of world total under alternative scenario I, but 74% of US and 14% of world total under alternative scenario II.
- **Of particular note,** we expect alternative scenario II would feature a "Japanese style adjustment," namely, despite a relatively sharp slowdown in headline GDP growth, consumption growth would remain very robust, as the consumption-GDP ratio ascends rapidly after the inflection point is crossed (exhibit 46).

Exhibit 45

China: Illustrative Forecast of Total Consumption under Base-case Scenario



Sources: CEIC, Morgan Stanley Research.

Exhibit 46

China: Illustrative Forecasts of Consumption

	Incremental Consumption as % of US		Incremental Consumption as % of World	
	2010	2020	2010	2020
Base-case	117%	210%	13%	20%
Alternative I	117%	123%	13%	12%
Alternative II	117%	246%	13%	24%

	Total Consumption as % of US		Total Consumption as % of World	
	2010	2020	2010	2020
Base-case	20%	66%	5%	12%
Alternative I	20%	49%	5%	9%
Alternative II	20%	74%	5%	14%

Source: Morgan Stanley Research.

A Golden Age for Consumption: Three Pillars

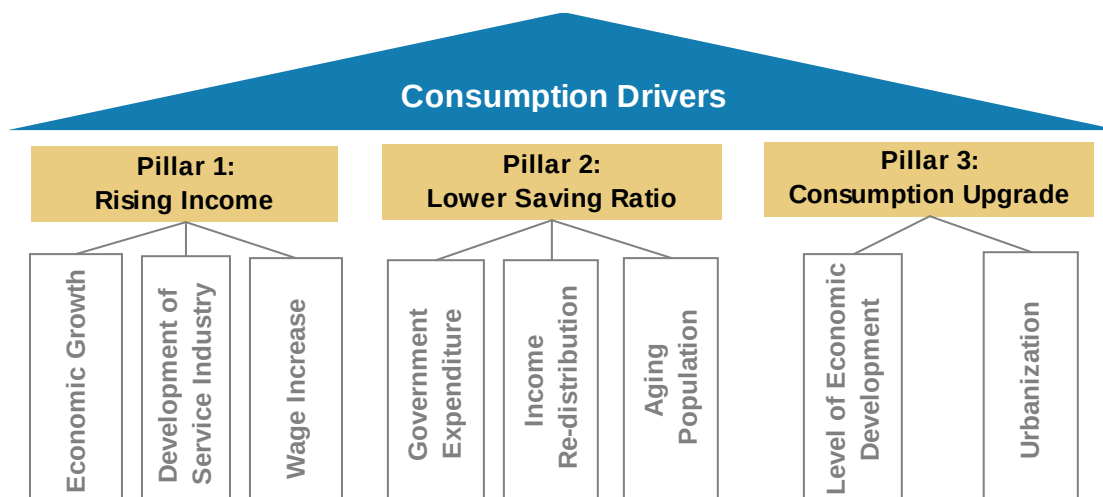
We believe that China is about to enter a golden age for consumption. We base this assertion on the historical experiences of Japan and Korea, which are most relevant in gauging China's long-term outlook, and on the the three scenarios, which are useful in illustrating the potential size of aggregate consumption in China through 2020. The key remaining question is how this golden age for consumption will materialize. More specifically, what are the potential drivers for consumption in practice?

Such a golden age would feature two major developments: strong expansion of consumption and a profound evolution of the consumption structure. To show how these two developments might unfold, we have identified eight drivers for consumption in China:

- Economic growth
- Development of service industries
- Wage increases
- Public expenditure
- Income redistribution
- Aging population
- Level of economic development
- Urbanization

We group these eight drivers under three pillars that underpin a golden age for consumption in China: rising income, lower savings ratio, and consumption upgrades (exhibit 47).

Exhibit 47

Three Pillars Underpin a Golden Age for Consumption through 2020

Source: Morgan Stanley Research.

Pillar I: Rising Income

The primary source of consumption growth stems from household income growth. Since compensation of labor is the most important source of income for the vast majority of Chinese households, household income in China is primarily a function of overall economic growth, concentration of labor in the economy, and wage rates. Looking ahead, we expect all three factors will provide robust support to rapid income growth over the next decade.

Economic Growth

Under the base case scenario, we believe the Chinese economy will still be able to maintain an average of 8% growth per annum over the next decade: This represents a slowdown of slightly more than 2 percentage points from the average growth achieved in the previous decade. We expect the deceleration in China's growth to be slower than that displayed by either Japan or Korea. Average CPI inflation over the next decade would be 3.5%, a significant acceleration from the average inflation rate of 1.9% over the previous decade. The higher inflation rate mainly reflects a labor market whose wages are increasingly normalized. At this pace of development, China's nominal GDP would triple from its current size, reaching Rmb103 trillion. Assuming an average 3% appreciation of the renminbi against the US dollar per annum, Chinese nominal GDP in US dollar terms would quadruple its current size, reaching US\$20 trillion by 2020.

Development of service industries

China's service sector is underdeveloped. Since typically the sector tends to provide a high number of jobs, this lack means that China's growth has, for the most part, been "jobless." Looking ahead, as the service sector develops, the labor intensity of the economy will increase, helping to boost household income growth.

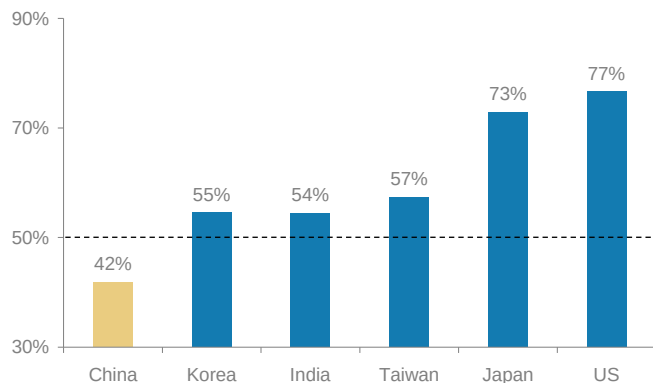
- Despite three decades of rapid development, the tertiary industry (service) represented just 42% of total GDP in 2009, not only far below its place in the advanced economies, such as the US and Japan, but also below developing peers, like India. Even compared to the levels in Japan 40 years ago and Korea 20 years ago, China's service sector is lagging (exhibit 48).
- The service sector is labor intensive, thus its lack of development helps to explain the jobless growth in China over the past two decades (exhibit 49).
- We expect service sector growth to outpace overall economic growth, so much so that the service sector as percent of GDP will keep rising, as indicated by the experiences of Japan and Korea (exhibit 48).

Development of the service sector will help to improve the labor intensity of the economy and boost the share aggregate labor compensation in the national income (exhibit 50).

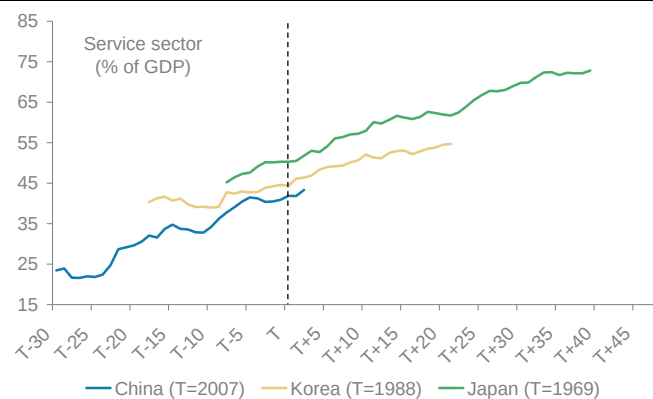
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Exhibit 48

Service Sector Is Underdeveloped in China...



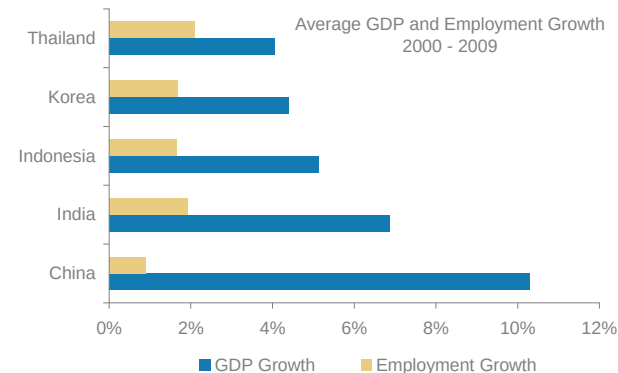
...Even Compared to Japan and Korea at Similar Stage of Development



Sources: CEIC, Morgan Stanley Research.

Exhibit 49

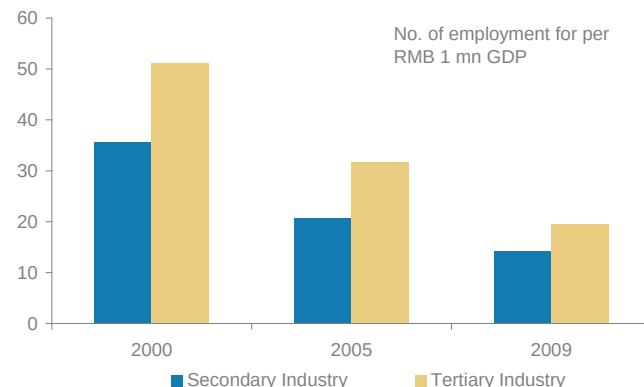
China's 'Jobless' Growth



Sources: CEIC, Morgan Stanley Research.

Exhibit 50

Development of Service Sector Is Job-Rich Growth



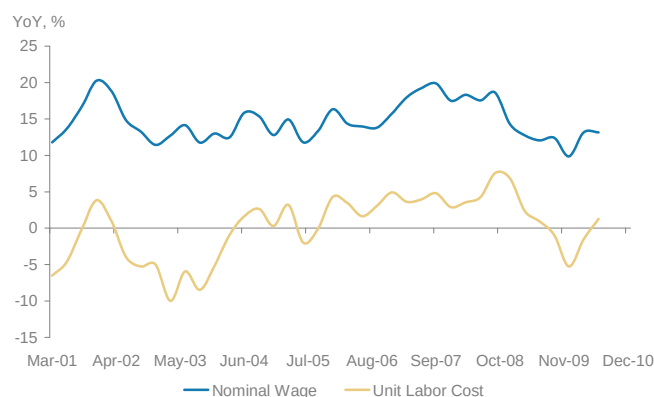
Source: CEIC, Morgan Stanley Research.

Wage increases

Labor market normalization has been under way for several years in China and should support sustained wage growth over the long run. The unit labor cost in the industrial sector has registered positive growth since 2004, which represents the beginning of the end of surplus labor supply in China (exhibit 51). Moreover, since then, seasonal labor shortages, especially in coastal areas (especially around Chinese New Year) have become the norm, instead of the exception.

Exhibit 51

Wage Growth in Industrial Sectors



Source: CEIC, Morgan Stanley Research.

As stated earlier in this report, we believe that labor supply in China through 2020 will remain abundant and unlikely to become a binding constraint that would cause overall economic growth to turn down sharply. That said, as labor market normalization is under way, wage growth in China is more likely to be strong—in line with labor productivity growth—instead of being depressed by the large pool of surplus labor, as has been the case over the past 30 years.

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On the policy front, a key policy priority under the twelfth five-year plan is to make sure that labor compensation increases in line with overall economic growth. To this end, a large majority of the provinces in China raised local minimum wages this year, and the authorities are reportedly considering a wage bill, which features a collective wage negotiation mechanism.

Pillar II: Lower Savings Ratio

A declining savings ratio would boost consumption growth as incomes rise, and we expect Chinese households' savings ratio to peak and even start to decline through 2020. The potential trends of three important factors over the coming decade will help bring about a lower household savings ratio, in our view. First, more government expenditure on public goods (express rail train, social housing) and services (education, healthcare) would help lower the precautionary savings of households. Second, income redistribution would help lower the overall savings ratio, as the savings ratio of high-income households tends to be higher than that of low-income households. Third, as population aging kicks in, the average savings ratio tends to decline.

Government expenditure

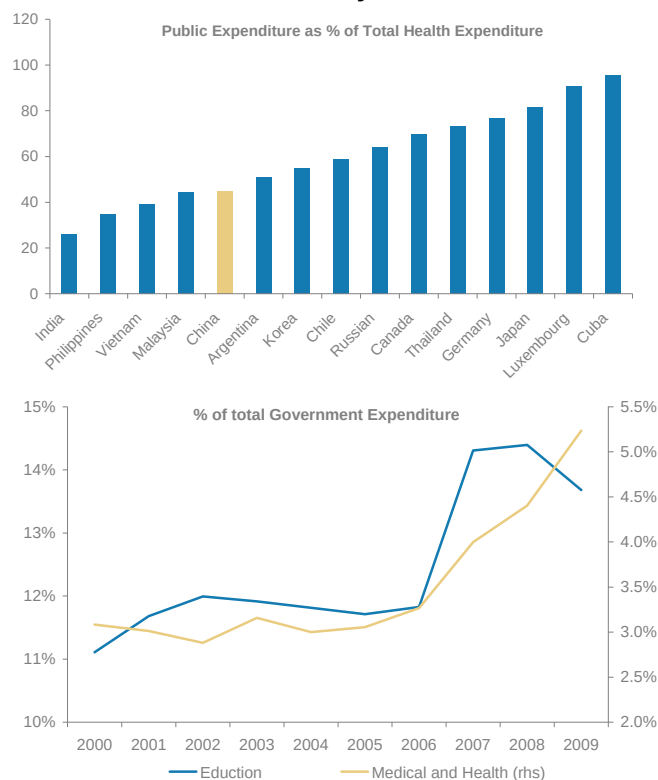
Government expenditure helps to reduce precautionary savings by households. The social welfare system has gone through profound changes during China's transition from planned to market economy. Under the planned economy, social welfare such as education, medical service, housing, and pension were largely free, so that personal expenditure on these things was almost negligible. The subsequent reform of the public welfare system has made education, medical services, and housing much more expensive than before, in some cases even unaffordable for the poor. Consequently, households have to save more to pay for those formerly public goods and services. Precautionary savings increased sharply as a result, crowding out consumption. For example, given the low penetration of commercial medical insurance in China, the portion of health expenditure financed by the government (45% in 2007) was very low in China, which requires households to set aside a considerable amount of additional savings for the health expenditures of family members (exhibit 52).

In recent years, the Chinese government has realized the importance and urgency of reforming its social welfare system if it wants to encourage consumption. Ambitious plans are being implemented in health care reform, aimed at full coverage of basic medical services; pension reform, to address the problems stemming from an aging population;

and social housing, to provide affordable housing to low-income households. As these policy initiatives come to be carried out, they would reduce uncertainty and boost consumer confidence, thereby lowering the current precautionary savings ratio.

Exhibit 52

China: Generally Low-level Public Expenditure in Healthcare Has Led to Policy Shift



Sources: World Bank, CEIC, Morgan Stanley Research.

Income redistribution

Income inequality in China has widened over the past decade. Addressing this has become, and will remain, a key policy priority.

- The Gini Coefficient, a measure of income inequality, has been rising since the debut of economic reform in the late 1970s. The indicator passed 40% in the 2000s, suggesting that income disparity in China is already quite serious (exhibit 53). While China stopped publishing the Gini Coefficient in 2005, we estimate that this indicator may have surpassed 50% by 2009, bringing it close to the levels of Latin American countries such as Brazil, Peru, and Argentina (exhibit 53).

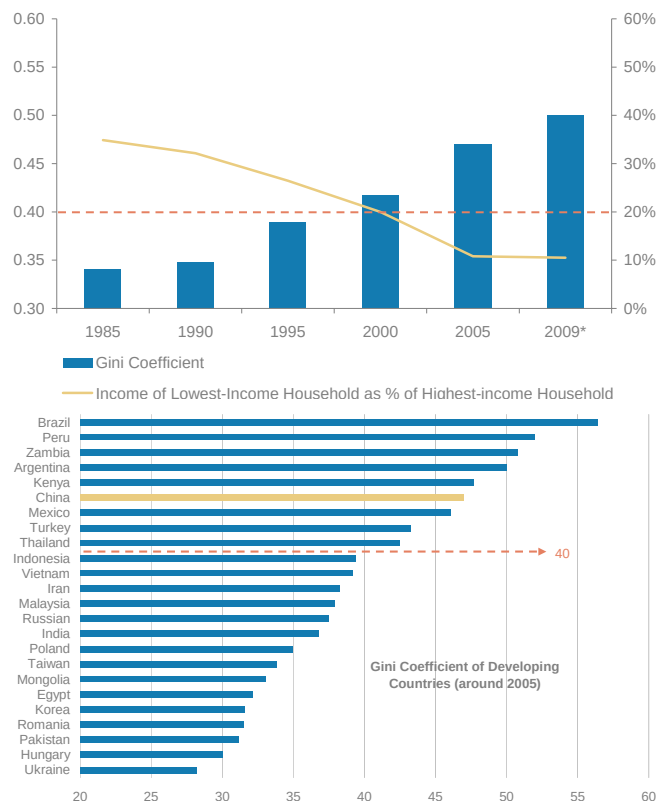
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Exhibit 53

Deterioration in Income Inequality



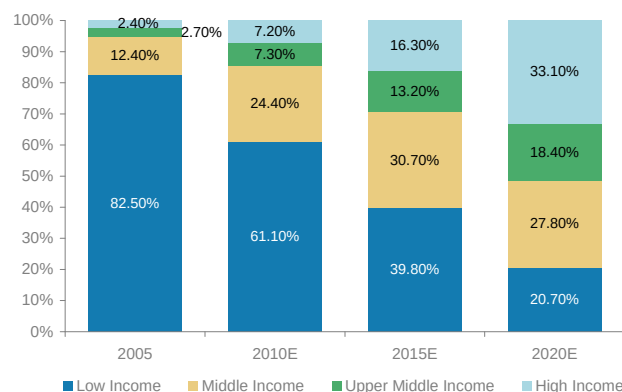
Sources: CEIC, Morgan Stanley Research * 2009 Gini Coefficient was an estimated figure.

- Since the propensity to consume is much higher in low-income households than in high-income ones, the rising income inequality has a negative impact on consumption growth. There is empirical evidence that every incremental increase of 1 percentage point of the Gini Coefficient would reduce the propensity toward consumption by 0.5-0.7 percentage points.
- Chinese authorities have realized the importance of income distribution to sustainable economic growth, especially as the underpinning of consumption growth. Besides strengthening the social safety net that targets the poor, we expect several tax measures to be implemented soon: The minimum threshold for personal income tax will be raised substantially to reduce the tax burden on middle- and low-income households; the marginal tax rate for the high-income bracket will be raised to quite a high level; and the real estate tax, or some special forms of property tax, could be implemented in 2011, as a means of taxing the stock of wealth (as opposed to flow of income) owned by the rich.

- These policy measures would help China achieve a more balanced income distribution featuring a large pool of middle-class households by 2020, according to the MS Asia/EM equity strategy team, led by strategist Jonathan Garner (exhibit 53).

Exhibit 53

China's Household Income Distribution



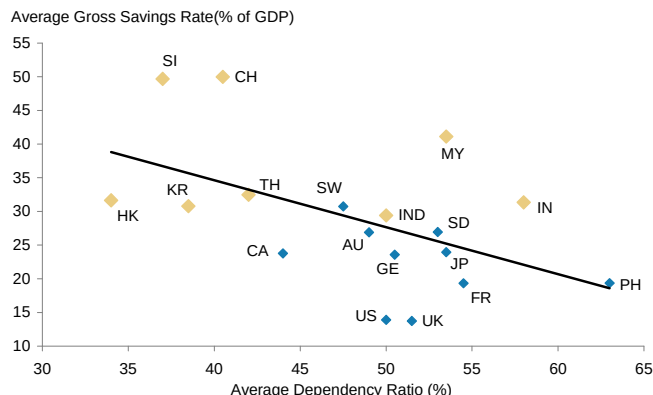
Sources: Euromonitor, Morgan Stanley EM/Asia Equity Strategy Team.

An aging population

The demographic profile is perhaps one of the most important factors in explaining the evolution of savings rates for an economy, especially for a developing or emerging market economy. In particular, cross-country data show a clear pattern that as a country's dependence ratio rises, its savings ratio tends to decline (exhibit 54).

Exhibit 54

High Dependence Ratio, Low Savings Rate



Source: Morgan Stanley Research.

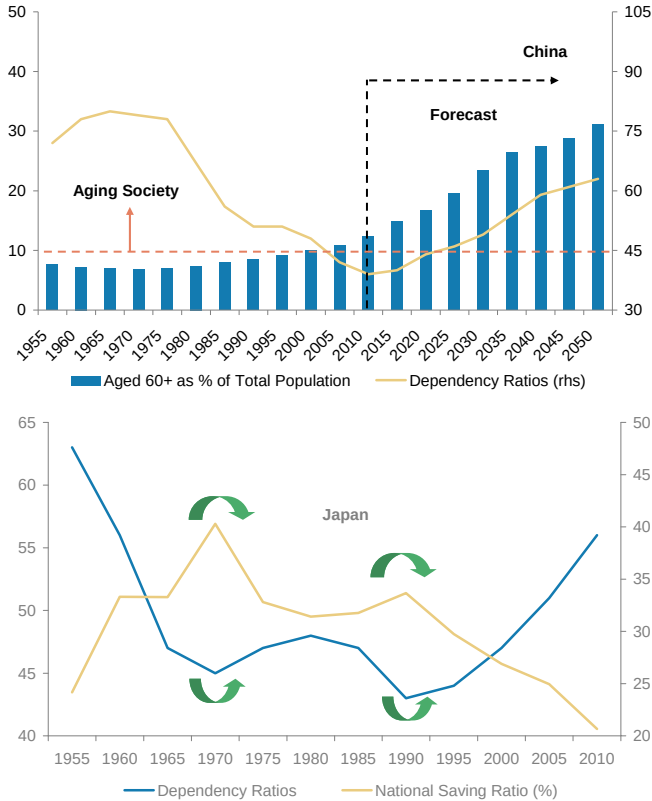
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Exhibit 55

Demographics and Inflection Point for Savings Rate



Sources: World Bank, CEIC, Morgan Stanley Research.

China formally became an “aging society” in 2005, according to the UN’s definition of such, where more than 10% of the total population is aged 60 or above (exhibit 55). According to the UN, China’s dependency ratio will bottom in 2010 and triple to 31.1% by 2050. The rising dependence ratio suggests that the overall savings ratio in China will peak out in the coming years and start to decline, perhaps as early as the second half of the next decade.

Pillar III: Consumption Upgrade

Recalling the beginning of this section, a golden age for consumption in China would feature two major developments: the rapid expansion of aggregate consumption and a profound and rapid change in the structure of consumption, or, a consumption upgrade. Consumers’ preferences and taste for different products and services typically reflect the level of economic development. In this regard, China is perhaps special in that as the economy develops, a massive structural shift is also taking place. The biggest shift in operation is rapid urbanization, which will have a tremendous impact on the

structure of consumption beyond the implications of the level of development.

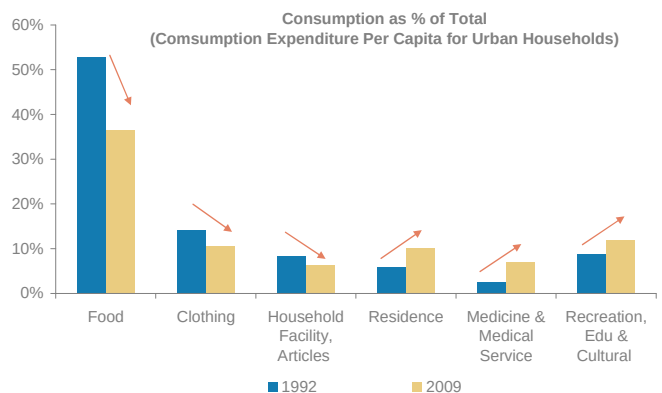
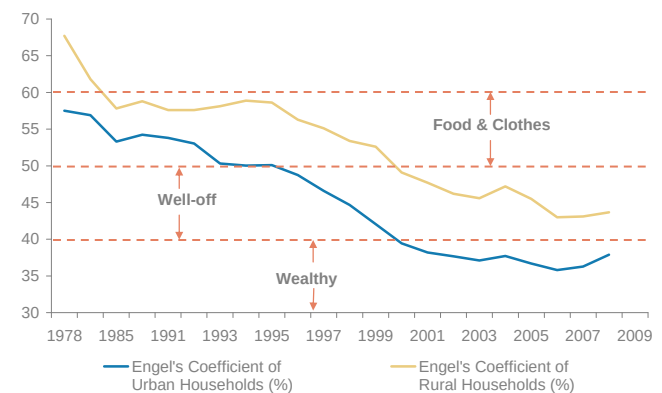
Level of economic development

Reflecting sustained economic growth, the living standards of Chinese households have improved substantially since the early 1980s.

- The Engel’s coefficient—the proportion of food expenses to consumption expenses—is an index used to evaluate a household’s standard of living. In China, this coefficient has registered significant decline for both urban and rural residents (exhibit 22).
- There has been broad-based consumption upgrade. Compared with the consumption structures in 1992 and 2009, the share of necessities like food, clothes, and home appliances declined, but the shares of residence, health care and recreation, education and culture—which represent more advanced-form of consumption—have risen markedly (exhibit 56).

Exhibit 56

Declining Engel’s Coefficient; Consumption Upgrading in Urban Sector



Sources: CEIC, Morgan Stanley Research.

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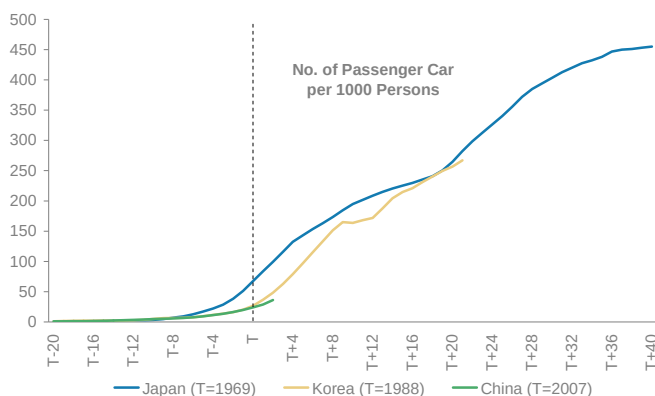
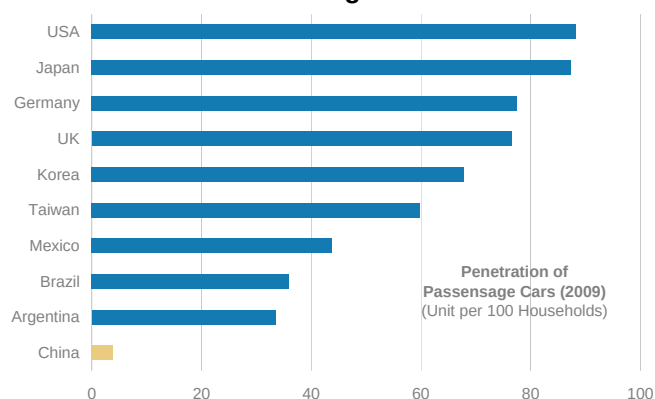
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- Compared with the consumption structure of more advanced economies, the consumption upgrade in China apparently has a long way to go. Passenger car penetration is a case in point (exhibit 57). During the 10 years after Japan and Korea reached the same level of development that China sees today, the number of passenger cars per 1,000 persons increased 3.5 times in Japan and 6.5 times in Korea.

Exhibit 57

A Long Way to Go with Consumption Upgrade in China: The Case of Passenger Cars



Sources: CEIC, Euromonitor, Morgan Stanley Research.

- We find that the consumption structure of China's urban households in 2007 was close to that of Japan in 1971 (exhibit 58). Specifically, food and clothing accounted for 37-38% of total private consumption, and housing was 15-16% for both urban China in 2007 and Japan in 1971. For health care, autos and telecoms, recreation, and education and culture, urban China in 2007 was only higher than Japan in 1971 by 1-2 percentage points.
- Assuming the evolution of China's urban consumption upgrade would follow Japan's pattern, we can expect that urban China's consumption structure will resemble that of Japan's in 1984: The share of daily necessities should

continue to fall, with food and clothing down 6.4 and 1.2 percentage points, respectively. The biggest winner would be health care, which will gain 2.6 percentage points, followed by housing (2.1 percentage points) and auto and telecom (2 percentage points). In other words, health care, housing, and auto and telecom will likely benefit most from China's consumption upgrade over the coming decade.

Exhibit 58

Consumption Upgrade: China vs. Japan

	Japan			Urban China	
	1971	1984	Changes	2007	2020
Food	29.2%	22.8%	-6.4%	29.8%	23.4%
Clothing	8.0%	6.8%	-1.2%	8.3%	7.1%
Residence	16.6%	18.7%	2.1%	14.6%	16.7%
Health Care	7.8%	10.5%	2.6%	8.7%	11.3%
Transport & Communi	8.0%	10.0%	2.0%	10.8%	12.8%
Recreation, Edu & Cultural	9.3%	9.7%	0.3%	10.7%	11.1%

Sources: CEIC, Morgan Stanley Research.

Urbanization

China has been experiencing rapid urbanization over the past two decades, with the urbanization ratio surging from 26.4% in 1990 to 46.6% in 2009, averaging 1% per year. Over 270 million rural residents moved into urban areas from 1990-2009, averaging 13 million per year. Nonetheless, China's urbanization remains low not only relative to developed economies like the US, Japan, and the UK, but also to Asian neighbors like Korea, Malaysia, and the Philippines (exhibit 59). China's urbanization is still roughly at the same level as that in the US a hundred years ago.

If the current pace of urbanization is maintained over the next decade, with the urbanization ratio rising 1% per year, China's urbanization rate should reach 58% in 2020. If, however, the average pace were to accelerate to 1.5% per year, which is likely, given that urbanization is now a priority with the Chinese government, we estimate that the urbanization rate would be able to reach 63% by 2020. This would imply that 12-20 million rural residents would be urbanized per year through 2020 (exhibit 59).

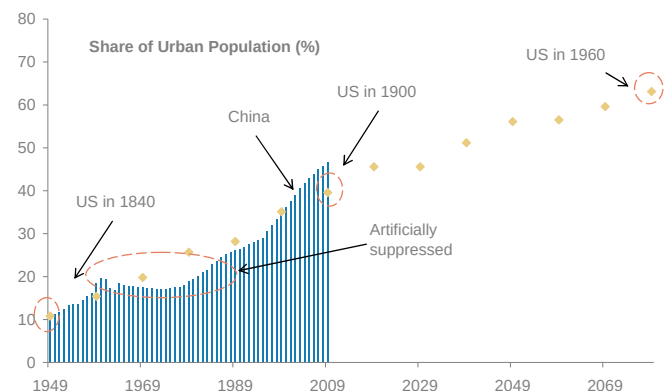
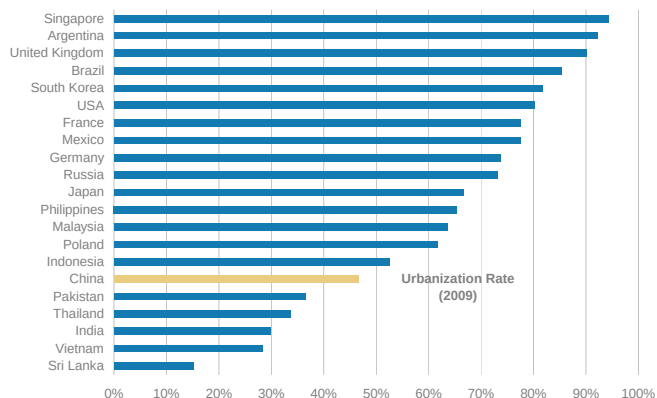
There is a large gap in income and consumption levels between urban and rural households in China. In 2009, rural household's income per capita was only 38% that of urban households, and consumption was only 51%. Housing aside, rural per-capita consumption of major goods and services was barely 50% that of urban households (exhibit 60). The penetration of major durable goods in rural areas lags far behind that in the urban sector (exhibit 26).

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Exhibit 59

Under-Urbanization in China



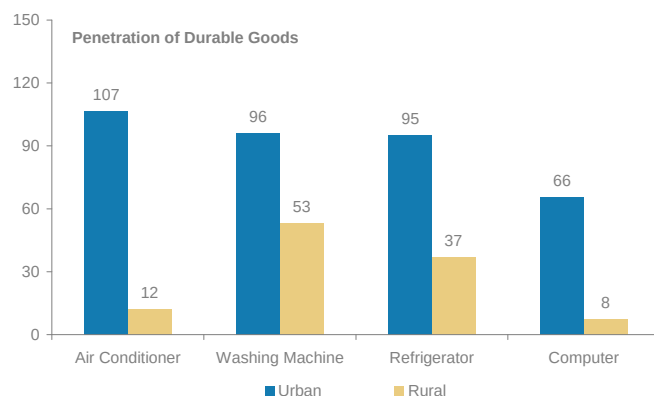
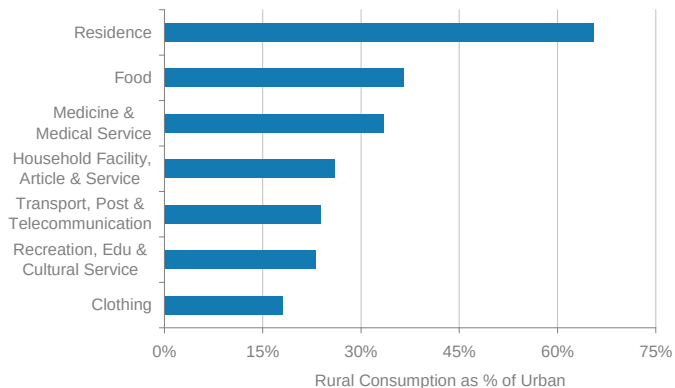
Sources: CEIC, Morgan Stanley Research.

We therefore expect that consumption upgrades in rural areas will be reflected in higher penetration levels of low-end durable goods (air conditioners, washing machines, and refrigerators), which have largely reached a saturation point in urban sector.

The still-sizeable difference between the development of urban areas and that of rural areas suggests that the evolution of China's consumption upgrades will differ from those of other advanced economies like Japan and Korea, where the producers of consumer goods had either to look for external demand or retrench production once domestic demand is satisfied. In China, while urban residents are seeking more advanced consumption in the form of high-end durable goods, and more services, the consumption upgrade in rural areas in the context of urbanization continues to generate strong demand for goods that have reached the saturation level in urban area. In this context, China's consumption upgrade is expected to benefit the entire industrial chain from low end to high end and from goods to services.

Exhibit 60

Consumption Gap in China: Urban vs. Rural



Sources: CEIC, Euromonitor, Morgan Stanley Research.

A Regional Perspective on the Golden Age

China is a big country, both in territory and in population. When it comes to levels of development, regional differences must also be taken into consideration as we make a case for the golden age of consumption in China (exhibit 27). We therefore dedicate a special section to discussing the potential dynamic trajectory of consumption across different regions in China.

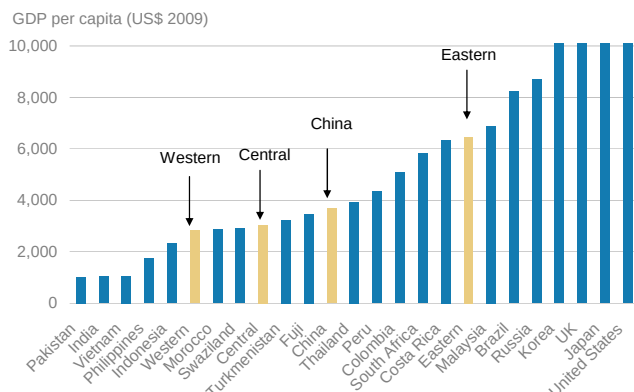
The economy of the eastern region took off much earlier than did that of the central and western regions. Besides the historical reasons for this, the eastern region was the first to benefit from the government's economic reforms and open-door policy. In terms of GDP per capita, the provinces of the eastern region averaged US\$6,468 in 2009, double the average of the central region (US\$3,043) and the western regions (US\$2,818) (exhibit 61).

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Exhibit 61

China's Regional Disparities in a Global Context

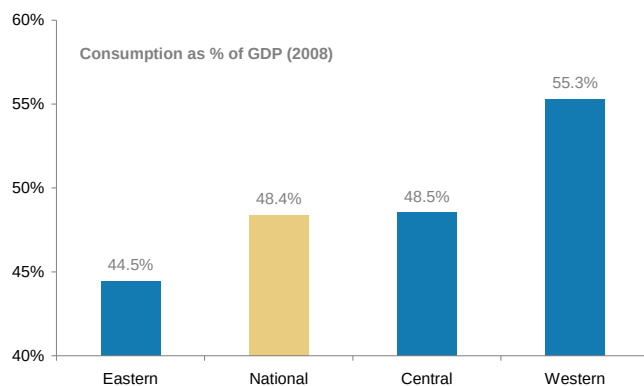


Sources: CEIC, Morgan Stanley Research.

Since the eastern region is the most developed of the three, it is perhaps natural to expect that the eastern region would have the greatest level of consumption (consumption-GDP ratio). In fact, the average consumption share of eastern provinces is lower than for the central and western regions (exhibit 62).

Exhibit 62

Myth: Low Level of Consumption in Eastern Region



Sources: CEIC, Morgan Stanley Research.

We explain the unusual relationship between the level of consumption and economic development by using a framework that describes the dynamic evolution of regional consumption (exhibit 63).

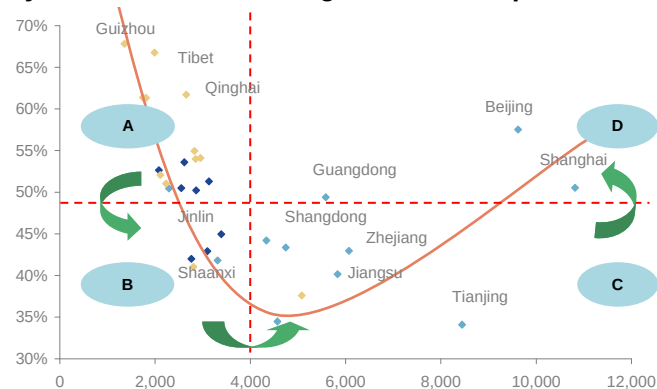
Construction of the framework

- If we map the ratio of consumption to GDP (Y axis) against the GDP per capita (X axis), it is interesting to observe that the plot points of 36 provinces form a conspicuous "U" curve (slightly tilting to the right-hand side).

- Taking this a step further, we may divide the space into four quadrants with two lines (vertical: US\$4,000 GDP per capita; horizontal: 48.6%, which is the national average consumption-to-GDP ratio).
- We then divide the space into four quadrants. Quadrants A and D represent a consumption-to-GDP ratio above national level while quadrants B and C are the opposite; quadrants A and D represent GDP per capita below US\$4,000, while quadrants B and C are the opposite.

Exhibit 63

Dynamic Evolution of Regional Consumption



Sources: CEIC, Morgan Stanley Research.

Initial observations

- The majority of western provinces (yellow spots) and some less-developed central provinces (dark blue spots) fall in quadrant A, which indicates a high level of consumption but low GDP per capita.
- Several central provinces land in quadrant B, which indicates a low level of consumption and low GDP per capita.
- The majority of eastern provinces fall in quadrant C, which indicates a low level of consumption but high GDP per capita.
- The three most developed provinces, Shanghai, Beijing, and Guangdong (all from eastern region), land in quadrant D, which indicates a high level of consumption and high GDP per capita.

The dynamic relationships among the four quadrants

- Quadrant A would suggest that inhabitants are in an over-consumption stage (high level of consumption but low income), while those in quadrant C are in an under-consumption stage (low level of consumption but high income). Quadrant B is the transition stage between quadrant A and quadrant C. Finally, quadrant D would be

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the mature stage, where we see a relatively balanced economic structure.

- The “over-consumption” stage here does not suggest runaway growth in consumption but rather the leanness of the contributions from investments and net exports. This also explains why some less-developed countries are found to have a higher level of consumption than do advanced industrial economies.
- For provinces in the over-consumption stage to excel, they would have to pursue investments and net exports very aggressively. As a result, the level of consumption may drop significantly, and these provinces would move from quadrant A to the transition stage in quadrant B, featuring both a low level of consumption and a low GDP per capita.
- Aggressive investments and exports could bring strong economic growth and a rising GDP per capita. But because the growth of investments and exports may still outpace consumption by a wide margin, the provinces may enter the “under-consumption” stage in quadrant C, featuring a low level of consumption but high GDP per capita.
- When marginal return of investments starts to peter out and export growth falters, the transition of economic growth would take place by returning to consumption. Then the economy may upgrade from quadrant C to quadrant D, where the level of consumption rises quickly.

Implications

- The consumption intensity of provinces in different regions should evolve along the “U” curve by following the trajectory of A to B to C to D.
- Over the next decade, not all provinces are likely to enter a golden age for consumption simultaneously. Provinces in quadrant C, in the “under-consumption stage”, will likely be among the first to enter a golden age for consumption, with the most rapid consumption upgrade. Provinces in quadrant B, in the “transition” stage, will likely also register strong consumption growth, but aggregate expansion at this stage is perhaps more profound than structural upgrade. Provinces in quadrant A are not likely to enter a golden age for consumption until perhaps the end of the next decade.
- It may take a longer time for China’s level of consumption to elevate (relative to GDP) when compared to the scenarios we explored with Japan or Korea, mostly because of the pronounced regional differences. This echoes the points we

made in the first section, “Not Whether But How Growth Will Decelerate.” This explanation for this lies in the relationships among the quadrants: While the level of consumption in the eastern provinces in quadrant C rises when moving into quadrant D, the level of consumption in the western provinces in quadrant A will decline when moving into quadrant B. These two offsetting trends point to a more extended period of rebalancing in China when compared to the historical situations in Japan or Korea.

Under-Consumption Overstated

While we share the consensus view that China’s consumption is relatively weak when compared with that of other economies, we dismiss the rather alarmist view that the Chinese economy is so seriously imbalanced as to pose a threat to economic stability in the short run. This is because we believe China’s official statistics substantially understate the true magnitude of consumption (especially the consumption of services) in China.

We addressed this issue in a report published in 2009 under the title “China Under-consumption Overstated” ([China Economics: China's Under-consumption Over-stated, September 13, 2009](#)). In that report, we reached the following conclusions:

- First, Chinese official statistics for personal consumption expenditure substantially underestimate its true magnitude and impact, primarily because of the underestimation of consumption of services—especially housing and health care—in China.
- Second, when we compare the consumption of non-services tradable goods between the US and China—which is, in our view, more relevant to assessing the impact on the rest of world—we see that the gap between the US and China is much smaller than suggested by overall consumption data (exhibit 64).
- Third, under a bottom-up approach, a like-for-like comparison of specific types of goods and services consumed by households in both countries indicates the magnitude of China’s personal consumption relative to that in the US could be much greater than is commonly perceived by the market (exhibit 64).

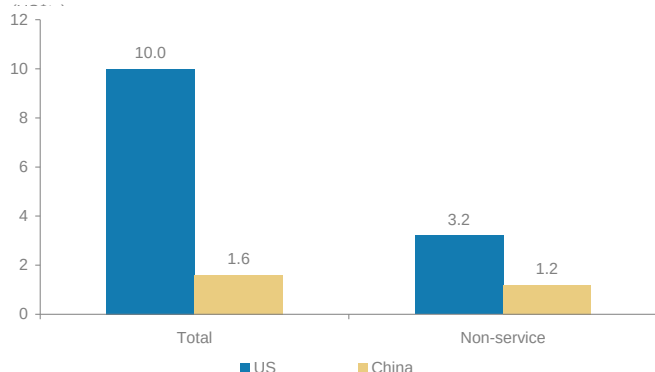
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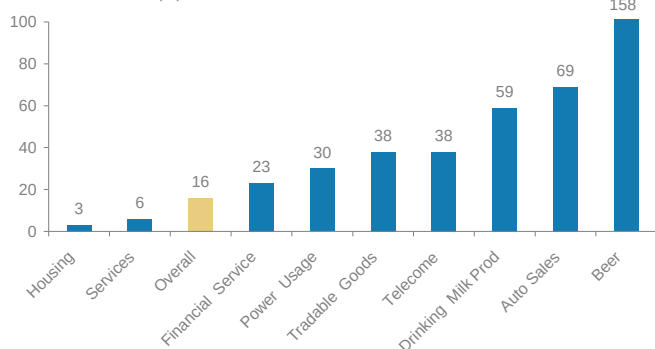
Exhibit 64

China: Under-Consumption Overstated

Personal Consumption in 2008



Comparison of Personal Consumption: Ratio of China over China (%)



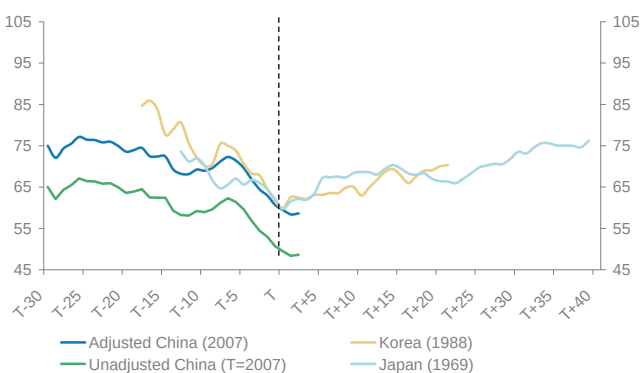
Source: Morgan Stanley Research.

We estimate China's private consumption-GDP ratio could be underestimated by at least 10 percentage points, or, personal consumption is underestimated by as much as 30%. In the same vein, we believe that China's investment-GDP ratio may have been overstated by the official statistics by as much as 10 percentage points.

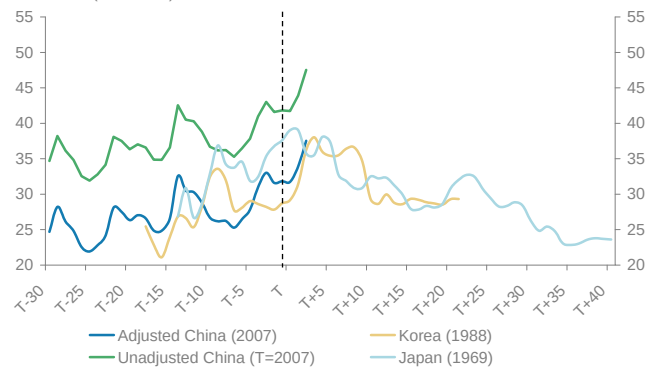
Exhibit 65

Adjusted Consumption and Investment Intensity

Consumption (% of GDP)



Investment (% of GDP)



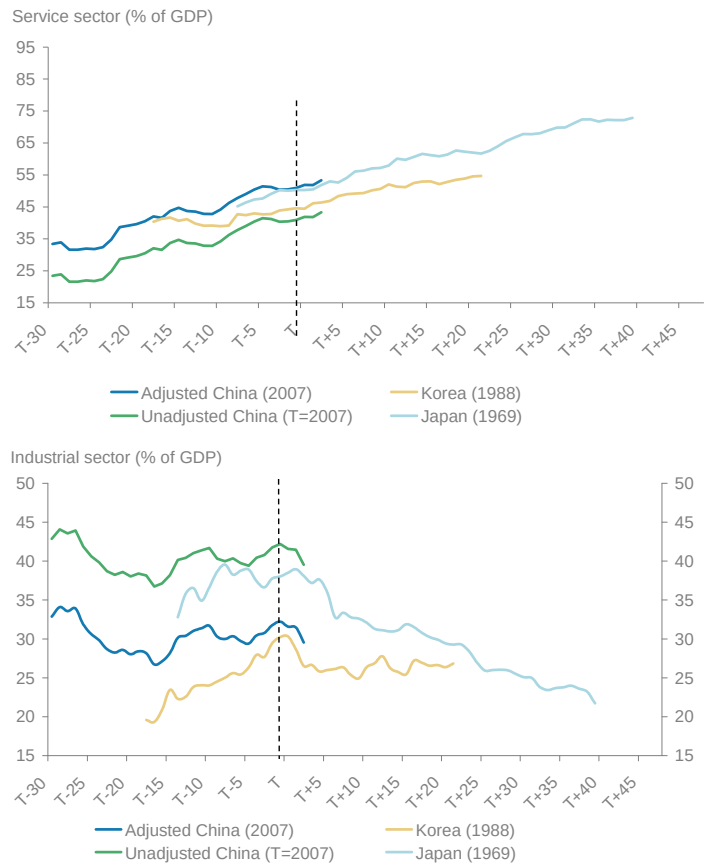
Sources: CEIC, Morgan Stanley Research.

If we do a simple exercise—adding 10 percentage points to consumption-GDP ratio while subtracting 10 percentage points from investment-GDP ratio—the trends of the adjusted ratios are amazingly similar to those of Japan and Korea at the same stage of development that China is in today (exhibit 65).

We believe that most of the error lies in an underestimation of the consumption of services in China, especially private consumption-related services (e.g., housing). If we do a similar adjustment to the production structure by adding 10 percentage points to the services-GDP ratio while subtracting 10 percentage points from the industrial sector-GDP ratio, the trends of adjusted ratios are also strikingly similar to those of Japan and Korea when the two economies were a similar stage of development (exhibit 67).

Exhibit 67

Adjusted Service and Industry Intensity



Sources: Company data, Morgan Stanley Research.

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The Americas
1585 Broadway
New York, NY 10036-8293
United States
Tel: +1 (1)212 761 4000

Europe
20 Bank Street, Canary Wharf
London E14 4AD
United Kingdom
Tel: +44 (0)20 7425 8000

Japan
4-20-3 Ebisu, Shibuya-ku
Tokyo 150-6008
Japan
Tel: +81 (0)3 5424 5000

Asia/Pacific
1 Austin Road West
Kowloon
Hong Kong
Tel: +852 2848 5200